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Cybersecurity Risks and Incidents Disclosure: A Literature Review*

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ABSTRACT

This paper reviews the expanding literature on the disclosure of cybersecurity risks and incidents. In contrast to prior reviews on cybersecurity, we focus specifically on disclosure and consider studies publicly available to the end of 2024. We classify the literature along three main dimensions: characteristics, determinants, and outcomes of cybersecurity disclosure. Within each dimension, we group studies that examine similar concepts to highlight areas where a critical mass of knowledge exists, as well as areas where research remains limited, findings are inconclusive, or contradictions persist. We conclude by outlining four broad research questions that warrant further investigation, with the goal of advancing our understanding of cybersecurity disclosure, its implications for organizations and stakeholders, the broader risk management landscape, and the related adversarial costs unique to this disclosure.

Keywords: cyber incidents, cyber risk, cybersecurity, disclosure, literature review

RISQUES LIÉS À LA CYBERSÉCURITÉ ET DIVULGATION DES INCIDENTS : EXAMEN DE LA LITTÉRATURE

RÉSUMÉ

La présente étude examine la littérature de plus en plus importante sur la divulgation des risques et des incidents liés à la cybersécurité. Contrairement aux examens antérieurs portant sur la cybersécurité, nous mettons l'accent sur la divulgation et prenons en compte les études qui étaient accessibles au public à la fin de 2024. Nous classons les études selon trois axes principaux : les caractéristiques, les déterminants et les conséquences des divulgations liées à la cybersécurité. Pour chacun de ces axes, nous regroupons les études portant sur des concepts similaires pour faire ressortir les domaines

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auxquels est associée une masse critique de connaissances, ainsi que ceux pour lesquels la recherche demeure limitée, les résultats sont peu concluants ou les contradictions persistent. Nous concluons en formulant quatre grandes questions de recherche qui méritent d'être approfondies, afin d'accroître nos connaissances sur les divulgations liées à la cybersécurité de même que leurs conséquences pour les organismes et les parties prenantes, le paysage global de la gestion du risque et les coûts adverses connexes propres à ce type de divulgations.

Mots-clés : cyberincidents, cyberrisques, cybersécurité, divulgation, examen de la littérature

1. INTRODUCTION

Because of their far-reaching implications, cybersecurity risks and incidents are a global concern, ranking among the top issues faced by organizations today (Kappelman et al., 2020). The scope and rise of cyber incidents underline that concern. For instance, the Equifax breach (May–July 2017) exposed the personal records of 147.9 million Americans, 15.2 million British citizens, and approximately 19,000 Canadians. The WannaCry ransomware attack in May 2017 affected more than 200,000 machines across 150 countries. The Yahoo data breach in 2013 compromised the information of over three billion Yahoo users. The LinkedIn breach in 2021 impacted 90% of its total user base, around 700 million user accounts (Hill & Swinhoe, 2022). A ransomware incident at Springhill Medical Center in 2019 that disrupted equipment responsible for relaying fetal heartbeat information to nurses was linked to the death of a baby, marking one of the first reported cases of a fatal ransomware attack (CBS News, 2021). In response, several laws, regulations, and guidelines call upon organizations to enhance cybersecurity risks and incidents disclosure. Specifically, one can refer to guidelines from the SEC (2011, 2018), to laws on data privacy security and data breach notification, and to industry-specific regulations.[1]

Within that context, adopting a critical perspective, this study reviews and synthesizes research on cybersecurity risks and incidents disclosure; identifies trends, patterns, and potential gaps in prior research; and offers suggestions for future research. Thus, we structure our review around "What does the literature on cybersecurity disclosure find?" using three lenses:

1. characteristics of cyber disclosure (the current state of the content and quality of cybersecurity risks and incidents disclosure, including its clarity, completeness, and accuracy),
2. determinants (drivers or barriers to disclose cybersecurity risks and incidents), and
3. outcomes (the extent to which cybersecurity disclosure is informative and effective in conveying insights to stakeholders about inherent risks).

1. The SEC's (2011, 2018) guidance uses the phrase "cybersecurity risks and incidents disclosure." Throughout the paper, we use this phrase interchangeably with "cybersecurity disclosure." "Disclosure" pertains to the communication of information in various forms, such as corporate reports, statements, other documents, and regulatory filings.

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We then put forward some research questions that may warrant further attention.

Our review leads to some observations. First, cybersecurity disclosure arises from a multifaceted spectrum of determinants spanning regulatory, public, and institutional pressures. Internal dynamics, including the interplay of board governance and firm characteristics, also contribute to the intricate tapestry of influences. Hence, debates persist regarding disclosure practices and the role of various determinants.

Second, our synthesis reveals that regulatory mandates, market dynamics, and cyber incident disclosure strategies collectively influence cybersecurity disclosure effectiveness. Moreover, there are broader implications from cybersecurity disclosure on stakeholder trust, firm performance, cost of capital, innovation, and spillover effects. Tensions arise between cybersecurity disclosure and market valuation, but also in market responses to cybersecurity incidents disclosure, including the reasons for such response (or lack thereof).

Third, there is a shortfall in the quality of cybersecurity disclosure when assessed against regulatory expectations, irrespective of regulatory context. These disclosures often use boilerplate language, lack uniqueness, have reduced readability, and tend to be litigation-aware, in contrast with the desired clarity and specificity. Irrespective of the jurisdiction, companies struggle to provide the necessary depth and completeness in their cyber incidents' disclosure. However, findings vary about the content and quality of cybersecurity disclosure linked to regulatory expectations, geographical variances, industry specifics, and the cyber incidents' impact.

Fourth, about half of the studies lack an explicit theoretical framework, while others apply a wide range of established frameworks. Our analysis suggests that theory-based and non-theory-based papers are complementary in providing insights into cybersecurity disclosure practices. Theoretical frameworks prevalent in financial disclosure literature enjoy widespread adoption, often yielding findings congruent with their theoretical premises. However, the literature has yet to recognize the fundamental disparities between the contexts in which financial disclosure and cybersecurity disclosure occur. Unlike the former, where disclosure primarily serves to inform stakeholders, cybersecurity disclosure occurs in an environment fraught with adversarial risks with potentially grave consequences. While financial disclosure may lead to proprietary, litigation, and political costs, cybersecurity disclosure brings about an additional type of cost—that of being targeted by adversaries with criminal intentions. Hence, we posit that financial disclosure frameworks may have limited explanatory power and call for the use of other theories and perspectives (e.g., game theory) that take the existence of an adversary into account.

Finally, we find that methodologically, most studies rely on quantitative textual analysis methods—specifically, word frequency assessments—or on event studies. In terms of disclosure outlets, 10-K filings are dominant, while other platforms such as conference calls with financial analysts or social media remain relatively underexplored. Most studies use cyber event data from the Privacy Rights Clearinghouse (PRC) and Audit Analytics.

The review offers the following contributions to research and practice. First, we add to the broader accounting disclosure literature. Considering the importance of

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cybersecurity and the rapid technological, regulatory, and legal developments underlying cybersecurity events, it is critical to better understand the disclosure of cybersecurity risks and incidents. Existing reviews either overlook cybersecurity *disclosure* altogether (Cram et al., 2019, 2023; Janvrin & Wang, 2022; Richardson et al., 2019; Spanos & Angelis, 2016) or focus on disclosure as a sub-theme with few papers (Haapamäki & Sihvonen, 2019; Walton et al., 2021). Further, our review extends their coverage as it encompasses five more years of published or yet unpublished studies.

Our focus on cybersecurity disclosure allows us to conduct a more in-depth review in which we uncover detailed themes surrounding the determinants that drive firms to disclose; the informativeness as well as the content and quality of disclosure; and the theoretical frameworks used within cybersecurity disclosure. By synthesizing and categorizing the literature, our study enhances the understanding of disclosure practices in response to evolving cyber threats and their implications. Our review not only addresses the limited coverage of cybersecurity disclosure in prior reviews but also provides a foundation for future research in this rapidly evolving field. In particular, we identify emerging trends and research gaps, as well as address methodological considerations.

Finally, our review considers the preventive and mitigation practices and measures used to address cybersecurity risks and incidents. Hence, the review provides practitioners with factors that motivate firms to disclose their cybersecurity risks, which may enable them to devise strategies that increase the overall level and transparency of cybersecurity risks and incidents reporting without increasing their vulnerability and exposure. Particularly, cybersecurity disclosure needs to balance standardization and firm specificity, to comply with both local and international regulations, and to accommodate the needs of various stakeholders. The limitations in the informativeness as well as the content and quality of cybersecurity disclosure call for better compliance with existing guidance as well as a need for enhanced monitoring from regulators.

2. BACKGROUND

Institutional and Regulatory Background

The National Initiative for Cybersecurity Careers and Studies (NICCS) (2022) defines "cybersecurity" as "the activity or process, ability or capability, or state whereby information and communications systems and the information contained therein are protected from and/or defended against damage, unauthorized use or modification, or exploitation" and "cyber incident" as "an occurrence that actually or potentially results in adverse consequences to (adverse effects on) (poses a threat to) an information system or the information that the system processes, stores, or transmits and that may require a response action to mitigate the consequences."

All organizations, whether regulated or not, are subject to legal duties pertaining to their corporate data such as the duty to protect the security of such data and the duty to disclose data breaches when they occur (Smedinghoff, 2015). This section summarizes the various

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disclosure obligations and public reporting requirements regarding cybersecurity risks and incidents.

Capital Markets Guidelines and Rules on Cybersecurity

In 2005, the SEC, through Regulation S-K, Item 305(c), requires that corporations include material risk information under Item 1A Risk Factor Disclosure of 10-K filings (SEC, 2005). However, the mandate does not explicitly mention the disclosure of cybersecurity risks and incidents.[2] The increasing number of cybersecurity attacks prompted the SEC to issue new disclosure guidance in 2011, this time including disclosure obligations relating to cybersecurity risks and incidents.[3]

The SEC requires a disclosure of cybersecurity risks and incidents in the following 10-K sections: (1) risk factors, (2) management discussion and analysis (MD&A) of financial condition and results of operations, (3) description of business, (4) legal proceedings, (5) financial statement disclosures, and (6) controls and procedures. The 2011 guidance mentions that "[while] no existing disclosure requirement explicitly refers to cybersecurity risks and incidents, a number of disclosure requirements may impose an obligation on registrants to disclose such risks and incidents." Furthermore, it adds that while firms "provide disclosure tailored to their particular circumstances and avoid generic 'boilerplate' disclosure, we reiterate that the federal securities laws do not require disclosure that itself would compromise a registrant's cybersecurity" (SEC, 2011). Hence, firms have discretion in deciding whether, what, and how much to disclose.

In 2018, the SEC issued an updated interpretive guidance to assist public companies in preparing and reporting their cybersecurity disclosure to investors (SEC, 2018). This interpretive guidance, which reinforces and expands on the SEC's 2011 guidance, addresses two new topics "namely the importance of cybersecurity policies and procedures and the application of insider trading prohibitions in the cybersecurity context" (SEC, 2018). Overall, the SEC's 2011 guidance and its 2018 interpretive guidance not only call for greater awareness for cybersecurity disclosure but also highlight the SEC's expectations for such disclosure.

On July 26, 2023, the SEC adopted final rules (SEC, 2023) that mandate the disclosure of material cybersecurity incidents on Form 8-K and require periodic reporting on a registrant's cybersecurity risk management, strategy, and governance within their annual

2. This matter stayed the same even after SEC's modernization of Regulation S-K Items 101, 103, and 105 (SEC, 2020a, 2020b). Specifically, in 2020, the SEC amended the Risk Factors disclosures requirements to reflect a more principles-based approach by only requiring disclosure of "material" risk factors, and "consistent with this principles-based approach, we are not adding a specific requirement to disclose cybersecurity risk as recommended by a commenter" (SEC, 2020a).
3. Although SEC guidelines are "not rules, regulations, or statements," they have "the de facto effect of law" (Ferraro, 2013). Based on analysis of SEC comment letters to 50 registrants requesting additional information about their cyber incidents and the registrants' subsequent correspondence, Ferraro (2013) concludes that registrants do not have a choice but to disclose more information about cybersecurity attacks despite claims of immateriality.

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reports. These regulations are designed to ensure uniform and comparable disclosure of significant cybersecurity incidents by registrants. For such incidents, reporting standards are now established on Form 8-K, with the stipulation for providing detailed information within four business days. There are exceptions to this rule in cases where immediate disclosure would pose substantial risks to national security or public safety. These new rules also introduce Regulation S-K Item 106, which demands comprehensive descriptions of the risk assessment processes, the material impact of cybersecurity threats, the role of the board in overseeing cybersecurity, and management's responsibilities in risk management. Foreign private issuers are also required to make similar disclosure, using Form 6-K for material cybersecurity incidents and Form 20-F for reporting on cybersecurity risk management, strategy, and governance. This regulatory development enhances transparency and consistency in cybersecurity reporting, benefiting investors, companies, and the broader market.

Similarly, in Canada, the Canadian Securities Administrators (CSA) consistently emphasize the significance of cybersecurity risks and the implementation of robust cybersecurity measures to safeguard issuers and stakeholders (CSA, 2013). This commitment is demonstrated through the release of the CSA Multilateral Staff Notice 51-347 in January 2017, which is aimed at guiding listed firms to enhance transparency and to better inform stakeholders about cybersecurity risk (CSA, 2017). The CSA requires disclosure in four major categories: (1) reporting of cybersecurity risks, (2) the potential impacts of risks on their business, (3) governance and other measures taken to mitigate risk, and (4) cyber incidents. Although this guidance has a comply-or-explain approach and is not mandatory, it is the first official move by Canadian market regulators to encourage firms to integrate cybersecurity into their disclosure obligations.

Industry-Specific Regulations

In addition to the SEC, the Federal Trade Commission and other government agencies impose industry-specific data security regulations. Organizations in the health care sector, for example, must comply with the Health Insurance Portability and Accountability Act (HIPAA) and HIPAA Security Regulations, while organizations in the financial sector must comply with the Gramm-Leach-Bliley Act (GLB). Stringent data security regulations also apply to the federal government and to critical infrastructure sectors.

State-Level Information Security and Data Breach Notification Laws

Organizations are also subject to a variety of state laws that require the implementation of security measures to protect corporate data, such as the California Consumer Privacy Act, the Utah Consumer Privacy Act, Massachusetts data security regulations, and others. The legal obligation to disclose security breaches to the persons affected stems from data breach notification laws issued across 50 US states.[4] However, these laws differ in how they define data breaches and personal data, penalties, content of notifications, notice to

4. See National Conference of State Legislatures (2020) for a summary and Congressional Research Service (2019) for further information on breach-reporting requirements.

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consumers, and notices to a supervisory body of data breaches. Similarly, in Canada, the Personal Information Protection and Electronic Documents Act (PIPEDA) forms the cornerstone of data protection, mandating the reporting of cybersecurity breaches that could inflict significant harm on individuals. Furthermore, at the provincial level, Alberta, British Columbia, and Québec have implemented their own private sector privacy laws, including specific provisions addressing cybersecurity and incidents reporting.

Global Regulations

Although many countries around the world have passed data privacy laws (e.g., Australia and Japan), a far-reaching law is the 2018 European Union (EU) General Data Protection Regulation (GDPR), which applies to any organization that stores or processes personal data about EU residents regardless of its domicile. From a disclosure perspective, GDPR requires firms to disclose breaches "without undue delay to the data subject … essentially going public about a security incident" (Schmitz-Berndt & Schiffner, 2021, p. 102).

Professional Bodies

Many professional organizations issued policies, procedures, guidance, and frameworks related to cybersecurity risks. For example, the PCAOB and the Center for Audit Quality (CAQ) provide tools and resources to board of directors members in identifying potential cybersecurity risks (CAQ, 2018; PCAOB, 2018). In 2017, the AICPA issued a Cybersecurity Risk Management Framework that provides organizations with criteria to evaluate their cybersecurity controls as well as communicate and manage information about their cybersecurity risk to interested parties (AICPA, 2017a, 2017b). Moreover, the AICPA issued documents that provide tools to the board members on how to effectively discuss cybersecurity risks and disclosure with management and CPA firms (AICPA, 2018).

Other organizations offer guidance on cybersecurity. COSO provides guidance on managing cyber risks in a digital age (COSO, 2015). The guidance explains how companies can apply COSO's Enterprise Risk Management to manage cyber risks and protect organizations against cyberattacks. In addition, the Institute of Internal Auditors (IIA) has a practice guide that discusses the internal audit activity's role in cybersecurity and how to provide assurance over cybersecurity risks (IIA, 2016). The Institute of Risk Management (IRM) has an outstanding report and practical guidance, which explores risk management in the context of cyber risk (IRM, 2014a, 2014b). Other cyber-focused standards and IT governance frameworks that provide conceptual roadmaps include COBIT 2019, ISO/IEC 27002, and National Institute of Standards and Technology (NIST) Cybersecurity Framework (ISACA, 2021; ISO, 2013; NIST, 2018).

Prior Cybersecurity Literature Reviews

The extant literature on cybersecurity encompasses a variety of reviews that provide valuable insights from different disciplinary perspectives and focus areas. For example, Haapamäki and Sihvonen (2019) synthesize 39 theoretical and empirical cybersecurity studies published in accounting and auditing journals. Their review identifies five primary research themes, including cybersecurity and information sharing, cybersecurity

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investments, internal auditing and controls related to cybersecurity, disclosure of cybersecurity activities, and security threats and breaches. Similarly, Cram et al. (2023) conduct a review focused on cybersecurity research within the field of accounting information systems (AIS), analyzing 56 articles published across 11 AIS-oriented journals. They categorize their findings into four key research areas: cybersecurity risks and threats; cybersecurity controls; cybersecurity-related assurance; and cybersecurity breaches and their consequences.

In a broader interdisciplinary context, Walton et al. (2021) review 68 cybersecurity papers spanning accounting, information systems, computer science, and general business disciplines. Their review classifies papers into five main categories, encompassing cybersecurity risk disclosure; cybersecurity investment; information security governance; market response to cybersecurity incidents and spillover effects; and responses from managers, auditors, and customers, along with remedial strategies. Additionally, other reviews focus on specific aspects of cybersecurity, such as the stock market reaction to information security events (Spanos & Angelis, 2016), economic consequences (Richardson et al., 2019), cybersecurity policies (Cram et al., 2019), and frameworks exploring the influence of cybersecurity on accounting (Janvrin & Wang, 2022). The timeliness of the topic means the cyber literature is moving quickly, so we further contribute by including studies publicly available to the end of 2024.[5] For comparison, the two most recent reviews, Walton et al. (2021) and Cram et al. (2023), include work up to the end of 2019.

While these reviews offer valuable insights into various dimensions of cybersecurity, they primarily focus on aspects other than the disclosure of cybersecurity risks and incidents.[6] In contrast, our review specifically centers on the theme of cybersecurity disclosure, thus filling what we deem to be a critical gap in the literature and providing a comprehensive analysis of disclosure practices in response to evolving cyber threats. Our review offers an enhanced understanding of the determinants, outcomes, and characteristics of cybersecurity disclosure practices and highlights areas where more research is needed to inform practice.

3. REVIEW METHODOLOGY

We conduct a systematic literature review to comprehensively explore the topic of cybersecurity risks and incidents disclosure. Given the multidisciplinary nature of cybersecurity and the rapid development of this field of research, we cast a wide net in terms of publication outlets

5. Three 2024 studies were subsequently published in journal issues in 2025 (Gauch & Quick, 2025; Singh, 2025; Trinh et al., 2025).
6. The overlap between the 79 studies included in our sample and each of these reviews is relatively small: 6 studies overlap with Haapamäki and Sihvonen (2019) [39 studies in their sample], 9 studies overlap with Walton et al. (2021) [68 studies in their sample], 8 studies overlap with Janvrin and Wang et al. (2022) [153 studies in their sample], and 6 studies overlap with Cram et al. (2023) [56 studies in their sample]. Given the emerging state of the cyber literature, we purposefully cast a wide net in terms of journal outlets. Other reviews restrict their sampled studies to those published in accounting (Haapamäki & Sihvonen, 2019) or AIS journals (Cram et al., 2023). Walton et al. (2021) include a wide range of journal outlets, but unlike our review, their review does not include unpublished working papers.

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and also consider unpublished working papers. Guided by a structured methodology, including a transparent search strategy and stringent inclusion criteria, the systematic review ensures analytical impartiality and comprehensiveness. We follow the guidelines in Kitchenham and Charters (2007) on the three stages for conducting a systematic literature review: planning, conducting, and reporting, which involve many iterations.

To gather the main corpus of papers, we conduct a keyword search (within the title, abstract, and keywords of papers) in major digital sources, including Scopus, Business Source Ultimate, ABI/INFORM, and Web of Science using terms from Haapamäki and Sihvonen (2019) and Walton et al. (2021) adapted to our specific focus on disclosure: "disclosure" and ["cybersecurity" or "cyber" or "security threats" or "cyber threats" or "cyberattack" or "information breach" or "data breach" or "data theft" or "information security" or "data security" or "IT security" or "security breach"], first in October 2023, and then again in December 2024 to update our review. We retain papers written in English and published in well-reputed academic journals (i.e., ABDC ranking B or better). In addition, using the same keywords, we search the Social Science Research Network (SSRN) to identify relevant unpublished working papers. We carefully read the abstracts to assess whether the study is indeed focused on cybersecurity disclosure, which further delimits the set of relevant studies. Lastly, we also search the bibliography of relevant studies for any other cyber disclosure papers that are not already in our sample. We end up with 79 relevant studies, of which 13 are working papers (16%).

Panel A of Table 1 shows the distribution of the studies we review by discipline. About 32% of the studies are published in accounting or AIS journals, followed by about 30% of studies in information systems and technology journals. We also cover business/management (9%), finance/economics (8%), and law journals (5%). Panel B of Table 1 shows the distribution of the studies by publication year, or year of latest update in the public domain in the case of working papers. We notice an uptick in the number of cybersecurity disclosure studies since 2018, which is the year the SEC issued its second set of guidelines. The trend also correlates well with the overall trend in cybersecurity incidents. Cybersecurity disclosure has been attracting a lot of attention recently, with 19 studies in 2024 alone. Next, we synthesize the dimensions of cyber disclosure in these studies as well as their findings in terms of determinants and outcomes, while critically evaluating the theory, methods, and variable measurement.

4. CYBERSECURITY DISCLOSURE CHARACTERISTICS

Building on the broader disclosure literature in accounting, Table 2 categorizes cybersecurity disclosure studies along three established dimensions: (1) disclosure channels—where firms disclose cyber risks and incidents, (2) disclosure characteristics—how firms disclose cyber risks and incidents, and (3) disclosure timing—when firms disclose cyber risks and incidents. Additionally, we identify a fourth dimension unique to cybersecurity disclosure: the source of disclosure—that is, who discloses the cyberattack. The following sections discuss the specificities of cyber disclosure along these dimensions.

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TABLE 1
Descriptives on the studies included in the review

Panel A: Distribution of journals across disciplines

Discipline	Frequency	Percent
Accounting (including AIS)	26	32.91
Business/management	7	8.86
Finance/economics	6	7.59
Information systems/technology	23	29.11
Law	4	5.06
Working paper	13	16.46
Total	**79**	**100**

Panel B: Distribution of studies by year of publication

Year	Published study	Working paper	Total
2006	1		1
2010	1		1
2011	1		1
2013	1		1
2016		1	1
2017	2		2
2018	5		5
2019	6		6
2020	6		6
2021	5	2	7
2022	15	2	17
2023	10	2	12
2024	13	6	19
Total	**66**	**13**	**79**

Notes: This table shows descriptive data on the studies included in this review. Panel A presents the distribution of the sample by the disciplinary domain of the journal that published the study. Working papers are not assigned to any discipline. Panel B presents the distribution of the sample by the year of publication. For working papers, we consider the latest year of update and distribution in the public domain (e.g., posted on SSRN).

Disclosure Channel

Panel A of Table 2 shows the studies we review classified by the disclosure channel in which cyber information is disseminated. A total of 38 studies examine annual reports and 10-K filings, particularly risk factor disclosures in Item 1A, following the SEC's 2011 cybersecurity disclosure guidance. While some studies focus on banks (e.g., Skinner, 2019; Zhang, 2024), most analyze broad cross-sections of US listed firms. A few studies (Firoozi & Mohsni, 2024; Héroux & Fortin, 2024) explore cybersecurity disclosure across multiple corporate documents (i.e., annual report, annual information forms, proxy statements, MD&A), aiming to create comprehensive disclosure indices. However, an underexplored aspect is the consistency of disclosures across different channels.

Beyond regulatory filings, nonregulatory press releases (examined in six studies) have been well researched, whereas other disclosure channels have received limited attention. Two disclosure channels specific to cybersecurity include (1) direct notifications to affected parties via letters (examined in three studies) and (2) regulatory filings to state attorneys' offices (examined in one study). Additionally, only one study (Jamilov et al., 2023) investigates

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TABLE 2
Cybersecurity disclosure channel, characteristics, timing, and source

Panel A: Disclosure channel—*Where* do firms disclose cyber risks and incidents?

Characteristic	Number of papers	Papers	
Annual reports/10-K Item 1A Risk Factors	38	Gordon et al. (2006) Hilary et al. (2016) Morse et al. (2017) Ettredge et al. (2018) Li et al. (2018) Skinner (2019) (US banks) Gao et al. (2020) Jeyaraj and Zadeh (2020) Li et al. (2020) Ashraf (2021) Berkman et al. (2021) Barry et al. (2022) Calderon and Gao (2022a, 2022b) Gatzert and Schubert (2022) Jiang et al. (2022) Peng and Li (2022) Radu and Smaili (2022) Wang et al. (2022) Chen et al. (2023)	Firoozi and Mohsni (2023) Florackis et al. (2023) Lattanzio and Ma (2023) Mazumder and Hossain (2023) Smaili et al. (2023) Tsang et al. (2023) Wu et al. (2023) Alodat et al. (2024) Du et al. (2024) Elnahass et al. (2024) Firoozi and Mohsni (2024) Héroux and Fortin (2024) Jiang (2024) Nelson and Wang (2024) Singh (2025) Trinh et al. (2025) Vo and Pham (2024) Zhang (2024) (banks)
Other parts of the annual report/10-K (Item 1 Business, Item 7 MD&A, Item 8 Financial Statements) or quarterly filings/10-Q	6	Hilary et al. (2016) Morse et al. (2017) Gao et al. (2020) Juma'h and Alnsour (2021) Firoozi and Mohsni (2024) Héroux and Fortin (2024)	
20-F reports (international firms listed in the United States)	2	Barry et al. (2022) Calderon and Gao (2022b)	
8-K filings in the United States/material change reports in Canada/ continuous disclosing regime in Australia	5	Skinner (2019) (US banks) Héroux and Fortin (2020) D'Arcy and Basoglu (2022) Andrew, Baker, Sheehan, et al. (2023)	Gordon et al. (2024)
Nonregulatory press releases	6	Chai et al. (2011) Wang et al. (2013) Gay (2017)	Bose and Leung (2019) Héroux and Fortin (2020) Foerderer and Schuetz (2022)
Conference calls	1	Jamilov et al. (2023)	
Proxy statements/circulars (i.e., form DEF 14A in the United States)	5	Skinner (2019) (US banks) Héroux and Fortin (2020) Klein et al. (2022)	Firoozi and Mohsni (2024) Héroux and Fortin (2024)
Annual information forms (AIFs) in Canada	3	Héroux and Fortin (2020) Firoozi and Mohsni (2024)	Héroux and Fortin (2024)
Direct communication with affected parties (notification letters)	3	Gwebu et al. (2018) Jackson et al. (2019) Nikkhah and Grover (2022)	
Regulatory filings to state attorney's office	1	Lin and She (2024)	
Forward-looking statements	1	Gao et al. (2020)	
Social media	1	Rosati et al. (2019) (Twitter)	

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TABLE 2 (continued)

Panel B: Disclosure characteristics—*How* do firms disclose cyber risk and incidents?

Characteristic	Number of papers	Papers	
Indicator of disclosure	12	Hilary et al. (2016) Morse et al. (2017) Li et al. (2018, 2020) Gatzert and Schubert (2022) Klein et al. (2022)	Radu and Smaili (2022) Smaili et al. (2023) Wu et al. (2023) Alodat et al. (2024) Elnahass et al. (2024) Trinh et al. (2025)
Length (words, sentences, paragraphs)	28	Hilary et al. (2016) Berkman et al. (2018) Li et al. (2018) Jackson et al. (2019) Gao et al. (2020) Berkman et al. (2021) Calderon and Gao (2021) Barry et al. (2022) Calderon and Gao (2022a, 2022b) D'Arcy and Basoglu (2022) Jiang et al. (2022) Klein et al. (2022) Peng and Li (2022)	Radu and Smaili (2022) Chen et al. (2023) Florackis et al. (2023) Jamilov et al. (2023) Lattanzio and Ma (2023) Mazumder and Hossain (2023) Smaili et al. (2023) Elnahass et al. (2024) Jiang (2024) Lin and She (2024) Nelson and Wang (2024) Vo and Pham (2024) Singh (2025) Trinh et al. (2025)
Readability (e.g., Gunning Fog Index; Readability Consensus Index)	7	Jackson et al. (2019) Gao et al. (2020) Calderon and Gao (2021, 2022a, 2022b)	Tsang et al. (2023) Jiang (2024)
Sentiment (positive/negative tone or news, uncertainty, litigation, etc., as per Loughran and McDonald (2011)	6	Gay (2017) Berkman et al. (2021) Calderon and Gao (2022a, 2022b) Tsang et al. (2023) Elnahass et al. (2024)	
Content analysis based on disclosure index	6	Skinner (2019) (US banks) Héroux and Fortin (2020) Nikkhah and Grover (2022)	Firoozi and Mohsni (2023, 2024) Héroux and Fortin (2024)
Specificity (e.g., cyber protections such as insurance, director expertise, types of risk or specific risk factors, contingent liability, investments, IT software packages, defense mechanisms, description of materiality)	17	Gordon et al. (2006, 2010) Chai et al. (2011) Wang et al. (2013) Ettredge et al. (2018) Bose and Leung (2019) Gao et al. (2020) Jeyaraj and Zadeh (2020) Berkman et al. (2021) Eijkelenboom and Nieuwesteeg (2021) Juma'h and Alnsour (2021) Nieuwesteeg and Eijkelenboom (2022)	Peng and Li (2022) Andrew, Baker, Sheehan, et al. (2023) Havakhor et al. (2024) Jiang (2024) Lin and She (2024)

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TABLE 2 (continued)

Panel B: Disclosure characteristics—*How* do firms disclose cyber risk and incidents?			
Characteristic	Number of papers	Papers	
Number of disclosures/ announcements/ press releases related to a cyber incident	2	Foerderer and Schuetz (2022) Andrew, Baker, Sheehan, et al. (2023)	
Modification scores/ similarity or dissimilarity scores/boilerplate	5	Gordon et al. (2010) Ashraf (2021) Jeyaraj and Zadeh (2022) Peng and Li (2022)	Wang et al. (2022)
Number of numbers	3	Jackson et al. (2019)	Calderon and Gao (2022a, 2022b)
Topic modeling (e.g., LDA, LLM model)	4	Cheong et al. (2021) Jeyaraj et al. (2021)	Du et al. (2024) Zhang (2024)
Panel C: Disclosure timing—*When* do firms disclose cyber risks and incidents?			
Characteristic	Number of papers	Papers	
Disclosure delay/ strategic timing	4	Gay (2017) Foerderer and Schuetz (2022)	X. Huang et al. (2024) Lin and She (2024)
Regulatory disclosure deadline	1	Ashraf et al. (2022)	
Panel D: Source of disclosure—*Who* discloses the cyberattack?			
Characteristic	Number of papers	Papers	
Firm discloses versus external party discloses	1	Amir et al. (2018)	

Notes: LDA, Latent Dirichlet Allocation; LLM, Large Language Model.

cybersecurity disclosures in corporate conference calls—an important channel in financial disclosure research due to its role in facilitating interactions between financial analysts and corporate management. Given the suggestion by Fortin and Héroux (2023) that financial analysts may not care about cybersecurity disclosure (based on seven interviews with investment managers and financial analysts), future research could explore whether the lack of studies on this channel stems from analyst disinterest or challenges in analyzing unstructured conference call data. Furthermore, with the SEC's 2023 guidance mandating the disclosure of material cyber incidents in 8-K filings, we expect increased research attention on this channel (currently examined in five studies).

Disclosure Characteristics

Regarding how firms disclose cybersecurity information (Panel B of Table 2), 12 studies use simple indicator variables to determine whether disclosure occurs, while 28 studies quantify disclosure length, often measuring the number of words in paragraphs that contain cyber-related keywords. In addition, 17 studies assess cyber disclosure specificity by

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examining references to cyber protections such as insurance, director expertise, types of risk, or IT software packages. Interestingly, Juma'h and Alnsour (2021) investigate contingent liabilities related to cyberattacks in financial statements' notes, bridging the gap with financial reporting research. This type of disclosure brings cybersecurity squarely into the realm of financial reporting research, so we believe focusing on this feature would highlight accountants' value added on this topic.

Disclosure Timing

The timing of cybersecurity disclosures relates to managerial strategies for delaying or promptly reporting adverse events. Four studies explore this dimension (Panel C of Table 2), highlighting the inherent tension between informing stakeholders and mitigating vulnerabilities before further attacks occur. Research in this area could provide valuable insights into the trade-offs firms face in determining disclosure timeliness. Ashraf et al. (2022) examine regulatory deadlines for disclosure across different US states, finding that tighter deadlines accelerate disclosures but at the cost of reduced incident detail. Given the SEC's 2023 mandate requiring disclosure of material cyber incidents within four business days, further research is warranted on the strategic timing of disclosures, its determinants, and its implications, particularly as they relate to the level of materiality of the cyber incident.

Source of Disclosure

Panel D of Table 2 presents the last dimension of disclosure characteristics we use to categorize cyber disclosure studies. Unlike other types of corporate disclosures, cybersecurity incidents may be disclosed not only by firms but also by external parties. Amir et al. (2018) is the only study to date examining whether cyberattacks are disclosed by firms versus external sources. In many cases, external actors—such as hackers, customers noticing suspicious activity, or other third parties—may possess and disclose information before or independently of the affected firm. This characteristic differentiates cybersecurity disclosure from traditional corporate disclosure and presents opportunities for further research. For instance, future studies could analyze the role of social media in cyber incident disclosures, including posts by affected customers, suppliers, or even attackers. Such research would complement existing studies on corporate cybersecurity disclosures via social media (e.g., Rosati et al., 2019).

Takeaways

By categorizing cyber disclosure research along established corporate disclosure dimensions, we highlight areas of greater and lesser research focus. The way in which firms disclose cyber risks in the risk factors sections of annual reports is well researched; however, further research into certain aspects would help paint a more complete picture of how managers decide what goes and what does not go into cyber risk disclosures (e.g., similarity over time or across companies or boilerplate cyber disclosures). We emphasize the distinct characteristics of cybersecurity disclosure that set it apart from traditional corporate disclosure and warrant further investigation (i.e., the source of the data breach announcement and disclosure about cyber risks and incidents outside the traditional corporate documents).

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Future research exploring these features would provide deeper insights into the evolving landscape of cybersecurity reporting.

5. DETERMINANTS OF CYBERSECURITY DISCLOSURE

Table 3 presents a summary of reviewed papers that examine the determinants of cybersecurity disclosure. Overall, it appears that five broad categories of determinants drive cybersecurity disclosure: (1) regulations, (2) prior cyberattack characteristics, (3) corporate governance, (4) peer effects, and (5) firm characteristics (such as size or profitability). The summary reports the author(s), journal title, the research objective(s), and the main findings. We now review the evidence from each type of determinant.

Regulations

Panel A of Table 3 highlights nine studies that identify regulations or regulatory pressures as potential determinants of cyber disclosure. Such pressures include the advent of the Sarbanes-Oxley Act (SOX) (Gordon et al., 2006; Sebastian, 2022); the enactment of the SEC's cyber disclosure guidance in 2011 (Gao et al., 2020; Hilary et al., 2016; Jiang et al., 2022) and 2018 (Firoozi & Mohsni, 2023; Gao et al., 2020); the adoption of the GDPR by the EU (Klein et al., 2022); and the enactment of regulatory guidance by the CSA (Firoozi & Mohsni, 2023, 2024). Overall, the evidence suggests that regulatory pressures lead firms to enhance their cyber disclosure across several measures (e.g., presence and frequency of cyber keywords or cyber-related words, existence of cyber index, readability). Moreover, Ashraf et al. (2022) show that state-level regulations can reduce the delays in disclosing cyberattacks. However, Ashraf et al. (2022) also point out that the specifics of the regulatory pressures, such as disclosure deadline, may nevertheless introduce complexities in disclosure content and timeliness.

Overall, we can infer from this evidence that firms adjust their cyber disclosure to meet level expectations embedded in regulations or regulatory guidance.

Cyberattack Characteristics

Panel B of Table 3 summarizes 10 papers that examine how a prior cyberattack affects a firm's subsequent cyber disclosure. The characteristics of cyber incidents, such as severity, type of data involved, firm-internal causes, prior breach experiences, and breach-related market reactions, influence firms' disclosure decisions. Overall, it does appear that the advent of a cyberattack as well as its severity leads firms to enhance their cyber disclosure on several dimensions such as readability (Jackson et al., 2019), number of words or keywords (e.g., Chen et al., 2023; D'Arcy & Basoglu, 2022; Gao et al., 2020; Jackson et al., 2019; Jeyaraj & Zadeh, 2020; Peng & Li, 2022), and frequency of cyber risk disclosures (Jiang et al., 2022). Adopting a slightly different perspective based upon voluntary disclosure theory, Amir et al. (2018) find that firms may tend to withhold information about cyberattacks especially if they are severe, with such delayed disclosure leading to a greater negative stock market reaction once the breach is discovered. Extending the disclosure scope, Jiang (2024) finds that the greater the estimated probability of

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TABLE 3
Determinants of cybersecurity disclosure

Panel A: Regulations (9 papers)

Paper	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Gordon et al. (2006)	21,810 firm-years, of which 1,992 disclose information security–related activities; 2000–2001 and 2003–2004 (pre- and post-SOX)	Disclosure of information security activities in annual reports based on list of keywords	SOX adoption in 2002	+	Table 3, p. 517
Hilary et al. (2016)	59,689 firm-years, 2000–2015	Cyber risk disclosure (0/1) in either Item 1A or Item 7 of 10-K	SEC 2011 cyber disclosure guidance	+	Table 1, Model 1, p. 47
		Number of cyber-related words and expressions in Item 1A and Item 7 of 10-K		+	Table 1, Model 2, p. 47
Gao et al. (2020)	1,344 firm-years, United States, 2007–2018	Number of words in cyber risk disclosures in 10-K	SEC 2011 cyber disclosure guidance	+	Table 9, Model 1, p. 18
			SEC 2018 cyber disclosure guidance	+	
		Readability of cyber risk disclosures in 10-K (higher values mean less readable)	SEC 2011 cyber disclosure guidance	n.s.	Table 9, Model 2, p. 18
			SEC 2018 cyber disclosure guidance	n.s.	
		Litigious language in cyber risk disclosures in 10-K	SEC 2011 cyber disclosure guidance	+	Table 9, Model 3, p. 18
			SEC 2018 cyber disclosure guidance	n.s.	
Jiang et al. (2022)	419 breach events corresponding to 3,206 firm-years, 2005–2017, PRC and Audit Analytics	Frequency of cyber risk disclosure in 10-K Item 1A (number of cyber keywords divided by the natural log of total number of words in Item 1A)	SEC 2011 cyber disclosure guidance	+	Table 4, p. 162

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TABLE 3 (continued)

Panel A: Regulations (9 papers)

Paper	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Klein et al. (2022)	3,700 US firm-years, 2014–2015 (pre) and 2016–2017 (post) when GDPR was adopted in the EU and that **never** had a cyberattack between 2005 and 2014 as per PRC	Cyber awareness (0/1) if the proxy statement DEF 14A contains the keyword "cyber" at least once	Post GDPR adoption in the EU	+	Table 3, p. 213
		Number of times the keyword "cyber" appears in the proxy statement DEF 14A		+	
Ashraf et al. (2022)	240 data breach events from Audit Analytics, US listed firms, 2010–2020	Cyberattack disclosure delay	Regulatory disclosure deadline (indicator variable) in state-level regulations	–	Table 3, p. 208
Sebastian (2022)	12 survey respondents approached via LinkedIn	Cyber disclosures	What-if scenario of cyber reporting requirement in SOX	Positive	Not applicable
Firoozi and Mohsni (2023)	8 largest Canadian banks and 40 largest US banks, 2014–2020	Cyber disclosure index (manually coded; extent of topics covered)	Canadian regulatory guidelines adoption (CSA, 2017)	n.s.	Table 3, p. 464
			SEC 2018 cyber disclosure guidance	+	
Firoozi and Mohsni (2024)	840 firm-years of Canadian listed firms, 2014–2020	Index of cyber disclosure across several corporate documents	Post Canadian regulatory guidelines adoption (CSA, 2017)	+	Table 7, p. 47

Panel B: Cyberattack characteristics (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Amir et al. (2018)	Voluntary disclosure theory (Dye, 1985) on withholding bad news	276 incidents from 156 US publicly-listed firms, 2010–2015; data from Audit Analytics and VERIS Community Database (VCDB)	Firm decision to disclose the cyberattack (vs. withholding the information, which is coded as the firm definitely knew about the cyberattack but an external party disclosed the information first)	Severity of the attack based on Gemalto index of type of data breached, number of records stolen, source of the breach, whether hackers used the stolen data	–	Table 3, Model 1, p. 1190

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TABLE 3 (continued)

Panel B: Cyberattack characteristics (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
				Damage (estimate of the damage the cyberattack caused divided by the market value of equity at the beginning of the year)	−	Table 3, Model 3, p. 1190
				Market reaction to cyberattack CAR (−1;+3)	+	Table 3, Model 2, p. 1190
Jackson et al. (2019)	Impression management	512 data breach incidents from 281 US firms, 2012–2015; data breach notification letters obtained from the Office of the Attorney General websites from California, Iowa, Maryland, New Hampshire, and Vermont	Readability Number of words Average number of characters per word Word count after removing duplicates Numerical intensity	Breach Level Index developed by IT-Harvest and SafeNet (i.e., data breach severity index that includes number of breached records, type of data accessed, source of data breach, how the data were used)	+ + + + −	Table 5, p. 1286
Gao et al. (2020)	Agency theory	1,344 firm-years, United States, 2007–2018	Number of words in cyber risk disclosures in 10-K Readability of cyber risk disclosures in 10-K (higher values mean less readable)	Cyber incident	+ n.s.	Table 9, Model 1, p. 18 Table 9, Model 2, p. 18

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TABLE 3 (continued)

Panel B: Cyberattack characteristics (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
			Litigious language in cyber risk disclosures in 10-K		n.s.	Table 9, Model 3, p. 18
Jeyaraj and Zadeh (2020)	Institutional theory: normative and coercive pressures shape cyber disclosures	87 firms from S&P 100 in 2018	Cyber response extracted from 10-K report based on keywords related to the actions, countermeasures, and portfolios related to cybersecurity	Normative pressure, measured as the loading of keywords describing various professional standards in the organization's annual report	n.s.	Table 2, p. 369
				Coercive pressure, measured as the loading of keywords related to laws and regulations in the organization's annual report	+	
Jiang et al. (2022)	Voluntary disclosure theory	419 breach events corresponding to 3,206 firm-years, 2005–2017, PRC and Audit Analytics	Frequency of cyber risk disclosure in 10-K Item 1A (number of cyber keywords divided by the natural log of total number of words in Item 1A)	Data breach (0/1) for all firm-years after a breach versus before	+	Table 4, p. 162
				Number of data breaches by firm from beginning of sample to year of the observation	+	
D'Arcy and Basoglu (2022)	Legitimacy theory	678 breach announcements across 381 US firms, 2005–2018 PRC	Number of cyber-related words in 8-Ks	Breach announcement (0/1)	+	Table 4, Model 1, p. 791
				Public pressure (i.e., average Google search volume activity from the	+	Table 4, Model 2, p. 791

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TABLE 3 (continued)

Panel B: Cyberattack characteristics (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
				day of the breach announcement to 3 days after, standardized by the baseline daily Google search volume over the 90 days prior to the breach announcement)		
Peng and Li (2022)	Cyber incidents are major events which we would expect to influence how managers disclose cyber risk	38 cyber incidents in the United States between 2013 and 2016 involving public firms; extract 10-K filings before and after each incident (76 10-K filings in total); Item 1A of 10-K filings	Change in number of words in cyber risk disclosures after to before a cyber incident	Number of records breached	+	Table 3, p. 463
			Change in number of risk factors included in the cyber risk disclosures		+	
			Modification (similarity) score in cyber risk disclosures from before to after the cyber incident		n.s.	
			Change in number of words in cyber risk disclosures after to before a cyber incident	Breach Level Index (assigned by the Data Breach Database)	+	Table 4, p. 464
			Change in number of risk factors included in the cyber risk disclosures		+	
			Modification (similarity) score in cyber risk disclosures from before to after the cyber incident		n.s.	

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TABLE 3 (continued)

Panel B: Cyberattack characteristics (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Chen et al. (2023)	Economic theory and stakeholder theory	1,116 firm-years of US listed firms that suffered data breaches, 2005–2018	Number of keywords related to cyber in 10-K Item 1A	Cyber breach	+	Table 4, p. 213
Zhang (2024)	Exploratory/machine learning	500 bank-years of 50 largest US banks, 2011–2020	Topic modeling (LDA) of cyber risk reports produced by two major institutions (Ponemon Institute and Verizon). Apply the topics identified to bank's annual reports and compute overlap. Higher overlap interpreted as better quality cyber disclosures	Time trend Cumulative total of data breaches	+ +	Figure 8, p. 24 Figure 10, p. 27
Jiang (2024)	Exploratory/machine learning	83,377 firm-years, US listed firms, 2005–2019	10-K text-based cyber insurance disclosure	Estimated probability of a data breach	+	Table 2, Panel C, p. 98

Panel C: Corporate governance (8 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Radu and Smaili (2022)	Critical mass theory	300 firm-years, S&P/TSX 60, 2014–2018	Cyber disclosure in annual report indicator variable (0/1)	Percentage of women on the BoD	+	Table 6, p. 365
				Blau Index of heterogeneity	+	Table 7, p. 366
			Number of words in cyber disclosure in annual report	Percentage of women on the BoD	+	Table 6, p. 365
				Blau Index of heterogeneity	+	Table 7, p. 366

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TABLE 3 (continued)

Panel C: Corporate governance (8 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
				One woman on the BoD (0/1)	–	Table 8, p. 367
				Two women on the BoD (0/1)	n.s.	Table 8, p. 367
				Three+ women on the BoD (0/1)	+	Table 8, p. 367
			Number of paragraphs in cyber disclosure in annual report	Percentage of women on the BoD	+	Table 6, p. 365
				Blau Index of heterogeneity	+	Table 7, p. 366
				One woman on the BoD (0/1)	–	Table 8, p. 367
				Two women on the BoD (0/1)	n.s.	Table 8, p. 367
				Three+ women on the BoD (0/1)	+	Table 8, p. 367
Mazumder and Hossain (2023)	Agency theory and resource based theory	210 bank-years of 30 listed commercial banks in Bangladesh, 2014–2020	Extensiveness of cyber disclosure measured as number of cyber-related keywords appearing in annual reports	Board size	n.s.	Table 6, p. 230
				Board gender diversity (% women)	+	
				Board independence (%)	+	
Smaili et al. (2023)	Stakeholder theory combined with improved monitoring of management if the board has financial expertise	300 firm-years of the 60 largest Canadian listed firms, 2014–2018	Cyber disclosure in annual reports (indicator variable)	Board effectiveness	+	Table 5, p. 1064
			Length of cyber disclosure in annual reports (number of words)	Board independence (%)	+	Table 6, p. 1065
				Board size	n.s.	
				Board financial expertise	+	
Héroux and Fortin (2024)	Upper echelons theory (Hambrick & Mason, 1984) combined with signaling theory (signals of "strong" board)	Cross-section of 250 largest Canadian firms on the Toronto Stock Exchange, 2018	Index that measures the extent of cyber disclosure across several corporate documents	Board IT expertise (%)	n.s.	Table 7, p. 387
				Board average tenure	n.s.	
				Board independence (%)	n.s.	
				Board gender diversity (% women)	n.s.	
				Average director age	n.s.	

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TABLE 3 (continued)

Panel C: Corporate governance (8 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
				Indicator variable if a board committee is responsible for cybersecurity	+	
Lin and She (2024)	Costs to prompt disclosure related to stakeholder anxiety and revelation of security vulnerabilities	219 firm-reported cyber incidents from Audit Analytics Cybersecurity Database, US listed firms, 2010–2021	Length of cyber disclosure in incident reports submitted to the state attorney's office (number of words)	Insider trading (indicator variable if insiders are net sellers from Day −10 to Day −1 relative to the reported discovery date)	+	Table 7, Panel B, p. 42
			Disclosure of how the attacked firm undertook cybersecurity enhancement measures following the incident as reported in incident reports submitted to the state attorney's office (indicator variable)		+	
			Attacked firm discloses that is has assured normal business operations as reported in incident reports submitted to the state attorney's office (indicator variable)		+	
Trinh et al. (2025)	Upper echelons theory (Hambrick & Mason, 1984)	About 1,250 firm-years (depending on the model) of US listed firms, 2002–2020	Likelihood of cyber disclosure in 10-K reports	CEO has PhD	−	Table 4, p. 46
				CEO serves as director on other boards	+	
				CEO tenure	n.s.	
				CEO is married	n.s.	
				CEO age	n.s.	
			Length of cyber disclosure in 10-K reports	CEO has PhD	−	
				CEO serves as director on other boards	+	
				CEO tenure	−	

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TABLE 3 (continued)

Panel C: Corporate governance (8 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
				CEO is married	+	
				CEO age	+	
Alodat et al. (2024)	Stakeholder theory, agency theory, resource dependence theory	2,250 firm-year observations of FTSE 350 UK listed firms, 2011–2020	Cyber disclosure in annual report (0/1)	Board size	+	Table 4, p. 11
				Board independence	+	
				Board meetings frequency	+	
				Board gender diversity	n.s.	
Elnahass et al. (2024)	Upper echelons theory, resource dependence theory, oversight role in agency theory, board composition as signal in signaling theory, stakeholder theory	1,177 firm-year observations of S&P 100 US public firms, 2002–2020	Likelihood of cyber disclosure in 10-K reports based on keyword search	Female directors (0/1, percentage)	+	Table 2, p. 10
			Amount of cyber disclosure in 10-K reports (number of words, number of paragraphs)		+	Table 3, p. 11
			Conservative tone of cyber disclosure		+	Table 4, p. 12
			Litigious tone of cyber disclosure		+	
			Uncertain tone of cyber disclosure		+	

Panel D: Peer effects (5 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Gao et al. (2020)	Agency theory	1,344 firm-years, United States, 2007–2018	Number of words in cyber risk disclosures in 10-K	Cyber risk index based on the total yearly number of cyber incidents in the United States as per Statista	+	Table 9, Model 1, p. 18
			Readability of cyber risk disclosures in 10-K (higher values mean less readable)		n.s.	Table 9, Model 2, p. 18

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TABLE 3 (continued)

Panel D: Peer effects (5 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
			Litigious language in cyber risk disclosures in 10-K		n.s.	Table 9, Model 3, p. 18
Jeyaraj and Zadeh (2020)	Institutional theory: mimetic pressures shape cyber disclosures	87 firms from S&P 100 in 2018	Cyber response extracted from 10-K report based on keywords related to the actions, countermeasures, and portfolios related to cybersecurity	Mimetic pressure, measured as average of weights for 10-K cyber disclosure of three competitors	+	Table 2, p. 369
Berkman et al. (2021)	Exploratory	14,305 firm-years, Russell 3000 firms, 2012–2018	Change in number of words in cyber disclosures (excerpts) from 10-K	Peer breach (0/1)	+	Table 3, p. 51
			Change in number of cyber risk mitigation words or phrases disclosed in 10-K based on a dictionary		n.s.	
			Change in negative tone in cyber excerpts in 10-K		n.s.	
			Change in natural log of one plus the number of IT software packages words		n.s.	
Ashraf (2021)	Disclosure of negative news is either disclosure of realized bad news or disclosure of potential negative outcomes. SEC cyber guidance limits managerial discretion,	1,284 breach incidents from PRC, 41,457 non-breached US firm-years, 2005–2017	Unique cyber factors computed as one minus the cosine similarity between firm *i*'s cyber risk-related risk factor disclosure in 10-K reports in year *t* and the cyber risk-related risk	Peer breaches (0/1) as proxy for the materiality of the potential cyber risk (i.e., proxy for non-breached firms' exposure to cyber risk)	−	Table 4, p. 34

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TABLE 3 (continued)

Panel D: Peer effects (5 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
	which may improve or harm the informativeness of risk disclosures		factor disclosures by all other firms in year *t*			
			Indicator SEC 2011 words in cyber disclosures		+	Table 10, p. 40
			Unique cyber factors computed as one minus the cosine similarity between firm *i*'s cyber risk-related risk factor disclosure in 10-K reports in year *t* and the cyber risk-related risk factor disclosures by all other firms in year *t*	Peer breaches (0/1) after SEC 2011 cyber guidance	–	Table 4, p. 34
			Indicator SEC 2011 words in cyber disclosures		+	Table 10, p. 40
D'Arcy and Basoglu (2022)	Legitimacy theory	678 breach announcements across 381 US firms, 2005–2018 (PRC)	Number of cyber-related words in 8-Ks	Data breaches in other firms in the same industry	–	Table 4, Model 5, p. 791

Panel E: Firm characteristics and other determinants (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Gao et al. (2020)	Exploratory	1,344 firm-years, United States, 2007–2018	Number of words in cyber risk disclosures in 10-K	Revenue	+	Table 9, Model 1, p. 18
				Profitability (ROA)	n.s.	
				Intangible assets	–	
				Auditor change	n.s.	
				Executive change	n.s.	
			Readability of cyber risk disclosures in 10-K (higher values mean less readable)	Revenue	+	Table 9, Model 2, p. 18
				Profitability (ROA)	n.s.	
				Intangible assets	–	
				Auditor change	n.s.	

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TABLE 3 (continued)

Panel E: Firm characteristics and other determinants (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
				Executive change	–	
			Litigious language in cyber risk disclosures in 10-K	Revenue	–	Table 9, Model 3, p. 18
				Profitability (ROA)	n.s.	
				Intangible assets	n.s.	
				Auditor change	n.s.	
				Executive change	n.s.	
Barry et al. (2022)	Internalized approach (United States) versus externalized approach (China) to managing cybersecurity	15,389 firm-year observations (561 Chinese cross-listed firm-years, 14,848 US firm-years), 2012–2018	Length and relevance of cyber words in 10-K and 20-F disclosures based on dictionary, normalized by year and industry (measure based on Berkman et al., 2018)	Chinese firm cross-listed in the United States	–	Table 4, p. 9
Wu et al. (2023)	Remote workforce brings up new cyber risks for the firm	27,747 firm-years, United States, 2018–2021	Likelihood of cyber risk disclosure	Remote workforce hiring before the pandemic	+	Table 5, Model 3, p. 35
Calderon and Gao (2022b)	International firms listed in the Unites States try to counteract US investors' home bias by providing more/better disclosure	4,632 firm-years (193 firms, 2007–2018), of which 2,316 10-K filings and 2,316 20-F filings	Length of cyber risk disclosures in 10-K or 20-F	International firm compared to US firms in US stock markets (indicator variable takes the value one if international, and zero if US firm)	+	Table 5, p. 76
			Number of numbers in cyber risk disclosures in 10-K or 20-F		+	
			Readability of cyber risk disclosures in 10-K or 20-F		–	
			Proportion of litigious words in cyber risk disclosures in 10-K or 20-F		–	

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TABLE 3 (continued)

Panel E: Firm characteristics and other determinants (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
			Proportion of uncertain words in cyber risk disclosures in 10-K or 20-F		−	
Gatzert and Schubert (2022)	Exploratory	124 US banks and insurers for a total of 992 institution-year observations, 2011–2018	Disclosure of cyber risk management in annual reports based on keyword search (0/1 indicator variable)	Cyber disclosure level/ extensiveness in annual reports ("awareness") based on keyword search	+	Table 8, p. 742
				Financial performance	−	
				Overall risk awareness	+	
Foerderer and Schuetz (2022)	Agenda-setting theory (McCombs & Shaw 1972, 1993)	2,757 event calendar-days	Number of data breach announcements	Nonaccounting news pressure on day of cyberattack disclosure (busyness in the trade press)	+	Table 3, p. 7306
Jamilov et al. (2023)	Gravity model	About 70,000–100,000 firm-quarters (depending on model) from 13,000 firms in 85 countries, 2002–2021	Country-level aggregation of cyber risk measured based on textual analysis of firm quarterly earnings conference calls	Country size (real GDP)	+	Table 5, p. 56
				Equity investment from the United States into the destination country	+	

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TABLE 3 (continued)

Panel E: Firm characteristics and other determinants (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
				Debt holding from the United States into the destination country	n.s.	
				Equity investment from destination country into the United States	n.s.	
				Debt holding from destination country into the United States	n.s.	
				Trade flow	n.s.	
				Disruptive tech exposure	+	
				Geopolitical proximity (Häge, 2011)	+	
				Common law	+	
				Physical distance from the United States	n.s.	
				Common language with the United States (English)	n.s.	
Andrew, Baker, Sheehan, et al. (2023)	Case study: Stakeholders have an impact on whether a data breach could become material to the firm	40 cyberattack announcements under the Australian continuous disclosure regime from companies listed on the ASX, 2018–2021	Disclosure of breach materiality	Subsequent reactions from customers and business partners	Positive	

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TABLE 3 (continued)

Panel E: Firm characteristics and other determinants (10 papers)

Paper	Theoretical perspective	Sample and period	Cyber disclosure variable	Determinant variable or concept	Main result	Source
Calderon and Gao (2022a)	Exploratory	490 SEC comment letters and responses, 2004–2019, Audit Analytics	Length of cyber risk disclosure in 10-K	Receiving an SEC comment letter about cybersecurity	+	Table 6, p. 11
			Number of numbers in cyber risk disclosures in 10-K		+	
			Readability of cyber risk disclosures in 10-K		−	
			Uncertain language in cyber risk disclosures in 10-K		−	
			Litigious language in cyber risk disclosures in 10-K		n.s.	
Wang et al. (2022)	Compliance versus strategic behavior given possibility information may be used by hackers	52 cybersecurity comment letters from the SEC, 2011–2019	Modification (dissimilarity) score in Item 1A cyber risk disclosures in the year after compared to before receiving an SEC comment letter on cybersecurity	Receiving an SEC comment letter about cybersecurity	+	Table 4, p. 8
				Responding to an SEC comment letter about cybersecurity	+	Table 7, p. 9

Note: ASX, Australian Stock Exchange; BoD, board of directors; CAR, cumulative abnormal return; FTSE, Financial Times Stock Exchange; GDP, gross domestic product; n.s., not significant; TSX, Toronto Stock Exchange.

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a data breach, the more firms are likely to disclose about their cyber insurance. Hence, overall, while a firm's prior cyberattack experience may underpin its cyber disclosure, there may be incentives for some managers to withhold information about severe attacks, especially if they were unexpected.

Corporate Governance

Panel C of Table 3 presents the summary findings of eight studies that consider corporate governance as a potential determinant of cyber security. Effective board governance emerges as a critical determinant of cybersecurity disclosure transparency. The synthesis reveals the influence of board characteristics, such as expertise, gender diversity (Elnahass et al., 2024; Mazumder & Hossain, 2023; Radu & Smaili, 2022), and independence (Alodat et al., 2024; Mazumder & Hossain, 2023; Smaili et al., 2023). Other variables, such as directors' age or tenure, CEO age, or IT expertise, do not clearly emerge as determinants. It also appears that there is more extensive disclosure when insiders are net sellers of a firm's shares in the 10 days prior to the reported discovery date (Lin & She, 2024). Focusing on a firm's CEO, Trinh et al. (2025) report that firms are less likely to provide cyber disclosure in their 10-Ks and that such disclosure would be shorter when the CEO has a PhD but that serving as a director on other boards leads to an outcome in the opposite direction.

It must be pointed out that while the studies we review in Panel C all relate to corporate governance, they draw upon a wide range of theories to explain their findings: critical mass theory (Radu & Smaili, 2022), agency theory (Alodat et al., 2024; Mazumder & Hossain, 2023), stakeholder theory (Alodat et al., 2024; Smaili et al., 2023), and upper echelons theory (Héroux & Fortin, 2024; Trinh et al., 2025). Elnahass et al. (2024) rely on all these theories in addition to signaling theory and resource dependence to explain the relation between woman directors and cyber disclosure.

Peer Effects

In Panel D of Table 3, we summarize five papers that examine how events or actions at a firm's peers influence its cyber disclosure. There is some limited evidence of mimetism with peers' cyber disclosure leading to more cyber-related keywords emphasizing actions, countermeasures, etc. (Jeyaraj & Zadeh, 2020). On a related note, Ashraf (2021) finds that firms reduced the uniqueness of their cyber risk disclosure following a peer breach with the advent of the SEC's 2011 guidance. His argument is that such guidance reduced managerial discretion and led them to engage in coercive isomorphism rather than normative isomorphism. Moreover, breaches at peers (Berkman et al., 2021) as well as a higher cyber risk frequency (based on cyber incidents) (Gao et al., 2020) translate into enhanced disclosure in terms of cyber-related words in 10-Ks. By contrast, D'Arcy and Basoglu (2022) find that more data breaches at other firms in the same industry led to less cyber-related words in 8-Ks.

Firm Characteristics and Other Determinants

Panel E of Table 3 reviews the 10 papers that consider various firm characteristics as potential determinants of cyber disclosure. In light of the large number of determinant variables considered, it is difficult to draw definite conclusions from these papers.

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However, not surprisingly, it seems that firm size (Gao et al., 2020), international scope (Calderon & Gao, 2022b), and receipt of an SEC comment letter (Calderon & Gao, 2022a; Wang et al., 2022) all contribute to enhanced cyber disclosure. Again, a wide range of theoretical perspectives is being applied to explain the findings.

Takeaways

The papers we review in this section look at a wide range of cyber disclosure determinants that are either external to the firm (regulatory pressures, cyberattacks, peer spillover effects) or internal (corporate governance, firm characteristics). Moreover, many theories are put forward to explain results (e.g., agency theory, voluntary disclosure, upper echelons theory) but none appears to dominate. We believe that theories that more closely reflect the uniqueness of cybersecurity, such as the adversarial costs of criminal activity, may lead to more interesting insights into the determinants of cyber disclosure. As an example, a relevant study may be Andrew, Baker, and Huang (2023), which theoretically discusses the role of cyber disclosures from the lens of surveillance capitalism. A remaining question is the relative importance of each set of determinants in driving a firm's cyber disclosure. Many studies focus on one or a few dimensions, and it is thus difficult to delineate the effect of various determinants. One may suspect that some determinants dovetail one another and work in conjunction. For instance, Ashraf (2021) shows that regulatory guidance, managerial incentives, and peers' events all interplay to affect a firm's cyber disclosure. Further work is thus needed to explore such interactions. Moreover, the settings also differ across the studies as some studies focus on firms' disclosure after a cyberattack or on cyber risk disclosure in general.

6. OUTCOMES OF CYBERSECURITY DISCLOSURE

In this section, we synthesize and critically evaluate the evidence on outcomes related to cyber disclosure. The literature so far has focused on five main types of outcomes: (1) stock market reaction, (2) public firm value, (3) other stock and debt market outcomes, (4) nonprofessional investors' willingness to invest, and (5) probability of subsequent cyberattacks. We regard these sub-streams of literature as reasonably well-developed, with at least seven studies relevant to each category. Eight studies examine other, more diverse types of outcomes; we group these in an "other" category, but synthesizing their findings is more challenging given the lack of a critical mass of studies of each one outcome.

Stock Market Reaction

Some of the prior literature focuses strictly on how the stock market reacts to the announcement of cybersecurity events. That literature is reviewed elsewhere (e.g., Richardson et al., 2019; Spanos & Angelis, 2016) and does not consider *where*, *how*, or *when* cyber risks or incidents are disclosed or *who* discloses (i.e., the cyber disclosure characteristics), which is where we contribute. Panel A of Table 4 presents the 20 studies we identify that examine either short-term or long-term market reaction to cyber disclosure characteristics. In this stream of literature, stock market reaction proxies for whether investors care about the cyber disclosure. Since there are many studies in this category, we split our discussion into cyber incident disclosure followed by cyber risk disclosure.

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TABLE 4
Outcomes of cybersecurity disclosure

Panel A: Stock market reaction (20 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Gordon et al. (2010)	1,641 disclosing and 19,266 nondisclosing firm-years; information security risk factor disclosure, 2000–2004	Indicator variable for a generic disclosure of information security in the annual report based on list of keywords	Annual stock market return, cumulated 9 months before fiscal year-end to 3 months after fiscal year-end	+	Table 7, Model 2, p. 581
		Indicator variable for a disclosure of proactive security activities only		+	Table 10, Panel B, Model 1, p. 585
		Indicator variable for a disclosure of potential security vulnerabilities only		+	Table 10, Panel B, Model 5, p. 585
		Indicator variable for a joint disclosure of proactive security activities and potential vulnerabilities		+	Table 10, Panel B, Model 5, p. 585
		Indicator variable for a joint disclosure of potential vulnerabilities and actual security breaches		n.s.	Table 10, Panel B, Model 5, p. 585
		Indicator variable for a joint disclosure of proactive security activities and actual security breaches		+	Table 10, Panel B, Model 5, p. 585
		Indicator variable for a joint disclosure of proactive security activities, potential vulnerabilities, and actual security breaches		+	Table 10, Panel B, Model 5, p. 585
Chai et al. (2011)	101 press releases announcing investments in IT security, from US public firms, 1997–2006	Disclosure of IT security investments	Stock market reaction CAR(−1;+1)	+	Table 5, p. 657
		Disclosure of IT security investments with commercial exploitation		+	Table 7, p. 658
		Disclosure of IT security investment without commercial exploitation		n.s.	
		Disclosure of IT security investment before SOX		n.s.	Table 8, p. 658
		Disclosure of IT security investment after SOX		+	

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TABLE 4 (continued)

Panel A: Stock market reaction (20 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Wang et al. (2013)	101 data breach announcements from 62 firms, Factiva press releases or news articles, merged with 10-K or 20-F risk factor disclosures immediately prior to the breach announcement; matched with 62 firms with information security risk factor disclosures but without breach announcements; 1997–2008	Cyber risk disclosure uses action-oriented terms (0/1)	Stock market reaction to breach announcement CAR(−1;+1)	+	Table 8, p. 214
		Disclosed cyber risk factors are realized subsequently		−	
		Percentage of disclosed cyber risk factors that are realized subsequently		−	
Hilary et al. (2016)	147 breach announcements, 2000–2015	Cyber risk disclosure (0/1) in either Item 1A or Item 7 of 10-K	Stock market return CAR(−1;+1)	n.s.	Table 3, p. 49
		Number of cyber-related words and expressions in Item 1A and Item 7 of 10-K		n.s.	
		Cyber risk disclosure (0/1) in either Item 1A or Item 7 of 10-K	Buy-and-hold 12-month abnormal return (long window returns)	n.s.	Table 5, p. 51
		Number of cyber-related words and expressions in Item 1A and Item 7 of 10-K		n.s.	
Morse et al. (2017)	295 firms disclosing cyber risk in 10-K or 10-Q filings between 2010 and 2012 (around SEC 2011 guidelines)	Voluntary cyber risk disclosure in 10-K or 10-Q	Stock market reaction CAR(−1;+1)	−	Table 6, p. 31
Gay (2017)	542 data breach events from DatalossDB.org, Databreaches.net, PHIprivacy.net, and PRC, 2005–2014; data on news events from Dow Jones News Service	Breaking news report released the same day as the data breach announcement	Abnormal stock return 30 days around data breach disclosure	−	Table 7, p. 102
		Abnormal positive news ratio released on the same day as the data breach announcement		+	
		Abnormal negative news ratio released on the same day as the data breach announcement		n.s.	

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TABLE 4 (continued)

Panel A: Stock market reaction (20 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Gwebu et al. (2018)	221 cyber incidents, 2000–2010 from PRC and DatalossDB.org; response strategies are coded based on the statements issued by the breached firm to its stakeholders	Accommodative response strategy to cyberattack by high reputation firms (included on *The Wall Street Journal*/ Harris Interactive Corporate Reputation list or *Fortune*'s Most Admired Companies)	Stock market reaction CAR(0;+1)	n.s.	Table 5, p. 705
		Moderate response strategy to cyberattack by high reputation firms		n.s.	
		Defensive response strategy to cyberattack by high reputation firms		n.s.	
		Image-renewal response strategy to cyberattack by high reputation firms		n.s.	
		Accommodative response strategy to cyberattack by low reputation firms		n.s.	Table 6, p. 706
		Moderate response strategy to cyberattack by low reputation firms		+	
		Defensive response strategy to cyberattack by low reputation firms		n.s.	
		Image-renewal response strategy to cyberattack by low reputation firms		+	
Amir et al. (2018)	276 incidents from 156 US publicly-listed firms, 2010–2015; data from Audit Analytics and VERIS Community Database (VCDB)	Firm decision to disclose cyberattack (vs. cyberattack is immaterial)	Stock return CAR(−1;+3)	n.s.	Table 6, Model 2, p. 1197
		Firm decision to withhold the cyberattack information (firm definitely knew about the cyberattack, but an external party disclosed) (vs. cyberattack is immaterial)		−	
		Firm decision to disclose cyberattack (vs. cyberattack is immaterial)	Stock return CAR(−1;+30)	−	Table 6, Model 4, p. 1197

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TABLE 4 (continued)

Panel A: Stock market reaction (20 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
		Firm decision to withhold the cyberattack information (firm definitely knew about the cyberattack, but an external party disclosed) (vs. cyberattack is immaterial)		−	
Rosati et al. (2019)	87 data breaches from 73 US publicly-listed firms, 2011–2014 based on PRC	Firm posts a tweet announcing the data breach	Stock market reaction CAR(0;+1)	−	Table 8, p. 467
		Firm posts a tweet announcing the data breach, and the firm has low coverage in the traditional newspaper media based on Lexis-Nexis		+	
Bose and Leung (2019)	526 press release announcements, 1995–2016, international sample	Disclosure of identity theft countermeasures via press release, Factiva search based on list of keywords	Stock market reaction CAR(−1;+1)	+	Table 2, p. 319
Juma'h and Alnsour (2021)	26 data breach events in the United States for which firms included information about the breach in their financial statements	Contingent liability disclosure related to cyberattack in the notes to financial statements	Stock market reaction (CAR −1;+1)	−	Table 4, p. 10
Wang et al. (2022)	52 cybersecurity comment letters from the SEC, 2011–2019	Receiving an SEC comment letter about cybersecurity	Stock market reaction (CAR −1;+1)	−	Table 6, p. 9
		Responding to an SEC comment letter about cybersecurity		−	Table 7, p. 9
Nikkhah and Grover (2022)	166 data breach announcements	Letter announcing cyber breach is "accommodative"	Stock market reaction (CAR −1;+1)	+	Table 11, p. 2180
		Letter announcing cyber breach is "corrective action only"		n.s.	
		Letter announcing cyber breach is "corrective action and apology"		+	
		Letter announcing cyber breach is "corrective action and compensation"		+	
		Letter announcing cyber breach is early		+	

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TABLE 4 (continued)

Panel A: Stock market reaction (20 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Ashraf et al. (2022)	240 data breach events from Audit Analytics, US listed firms, 2010–2020	Cyber disclosure delay	Stock market reaction (CAR −1;+1)	n.s.	Table 5, p. 210
Foerderer and Schuetz (2022)	208 cyberattack events, United States, 2008–2018	Nonaccounting news pressure on day of cyberattack disclosure (busyness in the trade press)	Stock market reaction (CAR −1;+1)	+	Table 9, p. 7314
Jamilov et al. (2023)	About 70,000–100,000 firm-quarters (depending on model) from 13,000 firms in 85 countries, 2001–2021	Text-based measure of cyber risk in quarterly earnings conference calls	Stock return	−	Table 6, p. 57
Chen et al. (2023)	1,116 firm-years of US listed firms that suffered data breaches, 2005–2018	Frequency of keywords related to cyber in 10-K Item 1A/extensiveness of cyber risk disclosure in 10-K Item 1A/more cyber risk disclosure following a breach	Stock market reaction to cyber risk disclosure in 10-K (CAR −1;+1)	n.s.	Table 6, p. 215
		Less cyber risk disclosure following a breach		−	
Florackis et al. (2023)	35,308 firm-years of US listed firms, 2007–2018	Text-based measure of cyber risk based on disclosure in 10-K Item 1A	Future stock market returns	+	Table 9, p. 375
Tsang et al. (2023)	378 observations of cyberattacked US firms, as per the PRC database, 2009–2018	Proportion of positive words in risk factor disclosures of cyber breached firms	Short-window stock market reaction (CAR −1;+1)	n.s.	Table 6, p. 172
		Proportion of negative words in risk factor disclosure of cyber breached firms		−	
		Number of uncertain sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of litigious sentiment words in risk factor disclosure of cyber breached firms		n.s.	

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TABLE 4 (continued)

Panel A: Stock market reaction (20 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
		Number of weak modal sentiment words in risk factor disclosure of cyber breached firms		+	
		Number of moderate modal sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of strong modal sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of constraining sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Readability of risk factor disclosure of cyber breached firms		n.s.	
Gordon et al. (2024)	Data breach events included in Audit Analytics Cybersecurity Database, 2005–2021	Location of cyber breach disclosure in 8-K vs. other documents	Short-window stock market reaction (CAR −1;+1)	−	Table 4, p. 8
			Post-announcement stock market reaction (CAR +2; +20)	+	

Panel B: Firm value (8 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Gordon et al. (2010)	1,641 disclosing and 19,266 non-disclosing firm-years; information security risk factor disclosure, 2000–2004; based on list of keywords	Indicator variable for a generic disclosure of information security in the annual report	Stock price 3 months after fiscal year-end	+	Table 5, Model 2, p. 577
		Indicator variable for a disclosure of proactive security activities only		+	Table 10, Panel B, Model 1, p. 585
		Indicator variable for a disclosure of potential security vulnerabilities only		+	Table 10, Panel B, Model 1, p. 585
		Indicator variable for a disclosure of actual security breaches only		n.s.	Table 10, Panel B, Model 1, p. 585

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TABLE 4 (continued)

Panel B: Firm value (8 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
		Indicator variable for a joint disclosure of proactive security activities and potential vulnerabilities		+	Table 10, Panel B, Model 1, p. 585
		Indicator variable for a joint disclosure of potential vulnerabilities and actual security breaches		n.s.	Table 10, Panel B, Model 1, p. 585
		Indicator variable for a joint disclosure of proactive security activities and actual security breaches		+	Table 10, Panel B, Model 1, p. 585
		Indicator variable for a joint disclosure of proactive security activities, potential vulnerabilities, and actual security breaches		+	Table 10, Panel B, Model 1, p. 585
Berkman et al. (2018)	9,677 firm-years of 2,264 US publicly-listed firms, 2012–2016	Cybersecurity awareness, based on length and relevance of cyber disclosure in 10-K	Share price 3 months after fiscal year-end	+	Table 4, p. 518
			Tobin's Q	+	Table 7, p. 521
Bose and Leung (2019)	526 press release announcements, 1995–2016, international sample	Disclosure of identity theft countermeasures via press release, Factiva search based on list of keywords	Long-term market value calendar-time portfolio analysis	+	Table 3, p. 321
Barry et al. (2022)	15,389 firm-year observations (561 Chinese cross-listed firm-years, 14,848 US firm-years), 2012–2018	Length and relevance of cyber words in 10-K and 20-F disclosures based on dictionary, normalized by year and industry (measure based on Berkman et al., 2018)	Share price at fiscal year-end	+	Table 5, p. 10
Gatzert and Schubert (2022)	124 US banks and insurers for a total of 992 institution-year observations, 2011–2018	Disclosure of cyber risk management in annual report	Firm market value (Tobin's Q)	+	Table 9, p. 744
Jamilov et al. (2023)	About 70,000–100,000 firm-quarters (depending on model) from 13,000 firms in 85 countries, 2001–2021	Text-based measure of cyber risk in quarterly earnings conference calls	Future market capitalization	−	Table 7, p. 58

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TABLE 4 (continued)

Panel B: Firm value (8 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Andrew, Baker, Sheehan, et al. (2023)	Case study: 40 cyberattack announcements under the Australian continuous disclosure regime from companies listed on the ASX, 2018–2021	Number of announcements/press releases for a cyber incident	Share price	–	Case study
Singh (2025)	475 firm-years of nonfinancial entities listed on the Bombay Stock Exchange, 2017–2023	Annual report text-based measure of cyber disclosure	Market capitalization as proxy for corporate reputation	+	Table 5, p. 10

Panel C: Other stock and debt market outcomes (7 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Gordon et al. (2010)	1,641 disclosing and 19,266 non-disclosing firm-years; information security risk factor disclosure, 2000–2004; based on list of keywords	Indicator variable for a generic/boilerplate disclosure of information security in the annual report	Bid-ask spread (average daily for 12 months starting 3 months after beginning of fiscal year)	–	Table 6, Model 1, p. 579
Berkman et al. (2021)	52,120 firm-years, United States	Number of words in cyber disclosures (excerpts) from 10-K Number of cyber risk mitigation words or phrases disclosed in 10-K based on a dictionary	Informed trading leading up to pre-earnings announcements (20-day window)—that is, abnormal stock volume, abnormal options volume, abnormal price range, abnormal intraday volatility	– –	Table 6, p. 59
Fortin and Héroux (2023)	Interviews: 7 interviews with investment managers, financial analysts, and chief information security officers at financial institutions	Cyber risk disclosure is mostly useless for sophisticated capital market participants due to it being perceived as boilerplate, unreliable, and incomparable			

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TABLE 4 (continued)

Panel C: Other stock and debt market outcomes (7 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Jamilov et al. (2023)	About 70,000–100,000 firm-quarters (depending on model) from 13,000 firms in 85 countries, 2001–2021	Text-based measure of cyber risk in quarterly earnings conference calls	Implied volatility	+	Table 87, p. 59
			Variance risk premium	+	
			Implied volatility slope	+	
Harris et al. (2023)	421 firm-years of US firms, 2010–2018	Length of cyber disclosure in 10-K filings (number of words); data from Gao et al. (2020)	Cost of debt	+	Table 4, p. 202
		Litigious terms in cyber disclosure in 10-K filings; data from Gao et al. (2020)		+	
		Readability of cyber disclosure in 10-K filings; data from Gao et al. (2020)		+	
Tsang et al. (2023)	378 observations of cyberattacked US firms, as per the PRC database, 2009–2018	Proportion of positive words in risk factor disclosures of cyber breached firms	Short-term trading volume (−1;+1)	−	Table 6, p. 172
		Proportion of negative words in risk factor disclosure of cyber breached firms		n.s.	
		Number of uncertain sentiment words in risk factor disclosure of cyber breached firms		−	
		Number of litigious sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of weak modal sentiment words in risk factor disclosure of cyber breached firms		+	
		Number of moderate modal sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of strong modal sentiment words in risk factor disclosure of cyber breached firms		+	
		Number of constraining sentiment words in risk factor disclosure of cyber breached firms		−	
		Readability of risk factor disclosure of cyber breached firms		n.s.	
		Proportion of positive words in risk factor disclosures of cyber breached firms	Short-term bid-ask spread (−1;+1)	−	
		Proportion of negative words in risk factor disclosure of cyber breached firms		n.s.	

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TABLE 4 (continued)

Panel C: Other stock and debt market outcomes (7 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
		Number of uncertain sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of litigious sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of weak modal sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of moderate modal sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of strong modal sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Number of constraining sentiment words in risk factor disclosure of cyber breached firms		n.s.	
		Readability of risk factor disclosure of cyber breached firms		n.s.	
Havakhor et al. (2024)	16,621 firm-years of US public firms, 2006–2018	Disclosure of cyber investments in SEC filings (indicator variable) instrumented by the firm receiving an SEC comment letter	Cost of equity Cost of debt	– –	Table 2, p. 36

Panel D: Nonprofessional investors—Experimental (8 papers)

Paper	Participants	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Frank et al. (2019)	547 participants on Amazon MTurk	Management assertion that its cyber controls are effective, as per AICPA (2017a) cyber reporting framework, in the absence of a prior cyberattack vs. when there has been a prior attack	Investment attractiveness	+	Table 3, Panel A, p. 194
		Cyber assurance report issued by audit firm that the firm's cyber controls are effective, as per AICPA (2017a) cyber reporting framework		+	Not tabulated, p. 195

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TABLE 4 (continued)

Panel D: Nonprofessional investors—Experimental (8 papers)

Paper	Participants	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
		Both management assertion and assurance report over effectiveness of cyber controls, as per AICPA (2017a) cyber reporting framework, in the absence of a prior attack vs. when there has been a prior attack		+	Table 5, Panel A, p. 196
		Cyber assurance report issued by audit firm that the firm's cyber controls are effective, as per AICPA (2017a) cyber reporting framework when a prior cyberattack occurred vs. without a prior cyberattack		+	Not tabulated, p. 195
Cheng and Walton (2019)	107 participants on Amazon MTurk	Company first discloses data breach versus external source	Investment judgment (attractiveness and likelihood)	+	Table 2, Panel D, p. 173
		Timely disclosure of cyber breach (gap between breach discovery and disclosure)		n.s.	
		Company first discloses data breach versus external source	Amount invested	+	Table 4, Panel D, p. 177
		Timely disclosure of cyber breach (gap between breach discovery and disclosure)		n.s.	
Kelton and Pennington (2020)	120 participants on Amazon MTurk	Cyber disclosures provided by non-breached firm prior to breach of industry peer	Investment contagion effect (investment attractiveness and likelihood of purchase)	n.s.	Table 4, p. 143
		Cyber disclosures provided by non-breached firm subsequent to breach of industry peer		−	Table 6, p. 145
Bansal and Axelton (2024)	313 participants, Amazon MTurk	Cyber disclosure specificity	Customer intentions	+	Figure 2, p. 84 (negative sign, but the dependent variable is defined as a decrease)
			Investor intentions	+	
			Employee intentions	+	
		Cyber disclosure verifiability	Customer intentions	+	
			Investor intentions	+	
			Employee intentions	+	

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TABLE 4 (continued)

Panel D: Nonprofessional investors—Experimental (8 papers)

Paper	Participants	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Demek and Kaplan (2023)	108 participants, Qualtrics	Firm discloses cyber risk management is a strategic initiative	Nonprofessional investor investment impression (favorable to unfavorable)	–	Table 2, p. 9
		CEO issues an apology post-breach		+	
		Firm discloses cyber risk management is a strategic initiative	Nonprofessional investor investment amount	–	Table 3, p. 10
		CEO issues an apology post-breach		+	
J. A. Huang and Murthy (2024)	151 participants on CloudResearch Connect	Disclosure of cyber insurance	Willingness to invest	+	Table 2, p. 11
		Disclosure of third-party cyber assurance		+	
		Disclosure of cyber self-assessment		+	
		Disclosure of cyber self-assessment with NIST		+	
Godewatta et al. (2024)	240 participants from Prolific	Positive tone of management disclosure of cyberattack	Investor willingness to invest (likelihood and attractiveness of investing in the company)	+	Table 1, p. 41
		Critical audit matter on the cybersecurity incident		n.s.	
		When investors have high IT knowledge, neutral tone of management cyber risk disclosure and critical audit matter about the cyber incident is present		+	Table 3, p. 42–43
		When investors have low IT knowledge, positive tone of management cyber risk disclosures and no critical audit matter about the cyber incident		+	
Gauch and Quick (2025)	100 UK participants from Prolific	Disclosure of assurance by an audit firm	Nonprofessional investor willingness to invest	+	Table 3, p. 16
		Disclosure of purchased cyber risk insurance		+	

Panel E: Subsequent cybersecurity attacks (8 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Wang et al. (2013)	101 data breach announcements from 62 firms, Factiva press releases or news articles, merged with 10-K or 20-F risk factor disclosures immediately prior to the breach	Cyber risk disclosure uses action-oriented terms (0/1)	Subsequent data breach	–	Table 5, p. 212

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TABLE 4 (continued)

Panel E: Subsequent cybersecurity attacks (8 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
	announcement; matched with 62 firms with information security risk factor disclosures but without breaches; 1997–2008				
Li et al. (2018)	26,626 firm-years, of which 291 firm-years with cyber incidents	Cyber disclosure (0/1) in 10-K filing (based on list of keywords)	Probability of subsequent cyber breach	+	Table 4, Panel A, p. 48
		Number of words in cyber risk disclosure in Item 1A of 10-K filing, divided by average number of words in individual risk factor disclosed in Item 1A		+	Table 4, Panel B, p. 48
Ettredge et al. (2018)	39,992 firm-years, of which 591 data breaches from 511 unique firms, US publicly-listed, 2007–2015	Disclosure of cyber defense mechanisms, based on the number of words related to cyber defense mechanisms in 10-K	Probability of subsequent cyber breach	–	Table 4, p. 576
		Disclosure of cyber vulnerability (i.e., number of words related to firms' cyber weaknesses in 10-K)		+	
Berkman et al. (2021)	14,305 firm-years Russell 3000 firms, 2012–2018	Number of words in cyber disclosures (excerpts) from 10-K	Data breach in the next year indicator variable (prediction of data breach)	–	Table 2, p. 49
		Number of cyber risk mitigation words or phrases disclosed in 10-K based on a dictionary		–	
		Negative tone in cyber excerpts in 10-K		–	
		Natural log of one plus the number of words related to IT software packages		n.s.	
Jamilov et al. (2023)	About 70,000–100,000 firm-quarters (depending on model) from 13,000 firms in 85 countries, 2001–2021	Text-based measure of cyber risk in quarterly earnings conference calls	Future cyberattacks	+	Table 3, p. 54
Florackis et al. (2023)	35,308 firm-years of US listed firms, 2007–2018	Text-based measure of cyber risk based on disclosure in 10-K Item 1A	Future cyberattacks	+	Table 6, p. 369
Du et al. (2024)	1,555 firm-years of US listed firms, 2018–2022; software data from Stackshare and the National Vulnerability Database	10-K Item 1A cyber disclosure extracted/summarized by LLM model	Future cybersecurity risk proxied by next year's vulnerability score of the software tools firms use	+	Table 3, p. 16
Jiang (2024)	83,377 firm-years of US listed firms, 2005–2019	10-K text-based cyber disclosure length	Data breach indicator variable	+	Untabulated
		10-K text-based cyber disclosure readability		+	

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TABLE 4 (continued)

Panel F: Other outcomes (8 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Gay (2017)	542 data breach events from DatalossDB.org, Databreaches.net, PHIprivacy.net, and PRC, 2005–2014; data on news events from Dow Jones News Service	Day prior to announcement of data breach	Number of positive news	+	Table 5, p. 99
			Number of negative news	n.s.	
		Day of announcement of data breach	Number of positive news	n.s.	
			Number of negative news	–	
Li et al. (2020)	12,238 firm-years, of which 131 with cyber incidents, 2010–2014; cyber incident data from Audit Analytics and PRC	Disclosure of cyber risks in Item 1A of 10-K (0/1) when the firm has suffered a cyber incident during the year	Audit fees (log)	n.s.	Table 5, p. 164
Calderon and Gao (2021)	7,450 US firm-years with cyber disclosures in 10-K reports, 2005–2018	Number of words in cyber disclosures (sentences)	Audit fees	+	Table 5, p. 33
		Gunning Fog readability		+	
		Number of litigious words in cyber disclosures		+	
Jamilov et al. (2023)	About 70,000–100,000 firm-quarters (depending on model) from 13,000 firms in 85 countries, 2001–2021	Text-based measure of cyber risk in quarterly earnings conference calls	Future firm performance	–	Table 7, p. 58
			Future cash flow from operations	–	
Nelson and Wang (2024)	4,405 firm-years, US listed firms with supplier data, 2010–2016	10-K cyber disclosure extensiveness from Berkman et al. (2018)	Change of sales to major customer (raw and as percentage of total sales of the supplier firm)	+	Table 3, p. 71
X. Huang et al. (2024)	326 firm-years of firms that disclose cyber breaches	PRC data breach, cyberattack disclosure delay (days from breach start to disclosure)	ROA	–	Table 1, p. 3
			CEO compensation	–	Table 3, p. 4
Vo and Pham (2024)	51,548 firm-years of US firms, 1998–2019	10-K text-based measure of cyber risk	Management guidance frequency	+	Table 2, p. 59
			Management guidance precision	–	Table 7, p. 64
			Management guidance accuracy	–	

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TABLE 4 (continued)

Panel F: Other outcomes (8 papers)

Paper	Sample and period	Cyber disclosure variable	Consequence/impact/prediction variable or construct	Main result	Source
Lattanzio and Ma (2023)	About 20,000 firm-years (depending on model) of US firms, 2000–2017	10-K text-based measure of cyber risk	R&D investments (intangible capital, asset intangibility, R&D to sales)	n.s.	Table 4, p. 11
			Patenting activity (count, citations)	+	
			Patenting time	−	
			Returns to innovation	−	

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In terms of *cyber incident disclosure*, Gwebu et al. (2018) and Nikkhah and Grover (2022) find somewhat inconsistent evidence on stock market reaction to the type of response strategy, with Gwebu et al. (2018) finding statistically insignificant short-term market reaction to response strategies such as defensive, accommodative, or image-renewal responses, while Nikkhah and Grover (2022) find some positive market reactions to letters that announce corrective actions, apologies, and compensations. Ashraf et al. (2022) find that cyber incident disclosure delay does not matter for how the stock market reacts, whereas Nikkhah and Grover (2022) find that more timely announcements lead to a positive short-term market reaction.

Disclosure channel seems to make a difference on how stock markets react to cyber incident disclosure. Breach announcements on social media (i.e., Twitter) lead to a negative market reaction, but if the firm has low coverage in the traditional newspaper media, that reaction will be positive (Rosati et al., 2019). The short-term market reaction to announcing a breach in 8-K regulatory filings is negative, but the longer-term post-announcement reaction (i.e., +2 to +20 days) is positive (Gordon et al., 2024). The findings on disclosure channel are particularly interesting in light of the SEC 2023 cyber disclosure guidance that material requires cyber incident disclosure in 8-K filings within four business days of discovery and in light of the increasing prevalence of corporate social media disclosure.

Gay (2017) and Foerderer and Schuetz (2022) find that news busyness on the day the firms announce a data breach can lead to a more muted or even positive stock market reaction, suggesting that firms may be able to "manage" stock market reaction to cyber incident disclosure by bundling the announcement with other news.

In terms of *cyber risk disclosure*, the provision and length of cyber risk factor disclosures in annual reports is generally not related to stock market returns (Chen et al., 2023; Hilary et al., 2016; Morse et al., 2017), and neither is the sentiment and readability of this disclosure (Tsang et al., 2023). The specificity of information risk factor disclosures in 10-K filings in early 2000 (Gordon et al., 2010)—which likely marks the initiation of this type of disclosure—and of IT security investments disclosed in press releases (Chai et al., 2011) is generally positively related to annual stock market returns. Interestingly, measures of cyber risk exposure based on the text in conference calls and in annual reports relate to stock market reaction in the opposite direction—negative relation for the former (Jamilov et al., 2023) and positive relation for the latter (Florackis et al., 2023).

Firm Value

Panel B of Table 4 shows the eight studies that examine the relation between cyber disclosure characteristics and firm value, where firm value is generally measured as share price, market capitalization, or Tobin's Q. Longer cyber disclosures in annual reports relate positively to firm value (Barry et al., 2022; Berkman et al., 2018; Singh, 2025). More specific cyber disclosures, in annual reports or press releases (e.g., disclosure of security vulnerabilities, identity theft countermeasures, or of other proactive security activities) also relate positively to firm value (Bose & Leung, 2019; Gordon et al., 2010). Interestingly however, conference call discussions about cyber risk (i.e., cyber risk exposure) relate negatively to future market capitalization (Jamilov et al., 2023), suggesting

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that when disclosure of cyber risk exposure occurs in disclosure channels that are time-constrained and interactive (i.e., limited time so topics tackled need to be selected), the cyber risks/exposure discussed may be more poignant, which further reflects negatively in corporate reputation and then into lower firm value. In addition, specifically related to cyber incident disclosure, a limited case study of 40 cyberattacks disclosed by Australian firms suggests that the more press releases a firm issues to disclose the attack, the lower the firm value (Andrew, Baker, Sheehan, et al., 2023).

The difference in findings for cyber disclosure in annual reports versus conference calls and press releases raises the possibility that the channel used for disclosing cyber risks is informative about how significant those risks are, and that the channel chosen to disclose cyber risks has predictive value for firm value.

Other Stock and Debt Market Outcomes

Panel C of Table 4 presents the seven studies that examine other stock and debt market outcomes of cyber disclosure, such as cost of equity, cost of debt, measures of information asymmetry (bid-ask spread, liquidity, trading volume, informed trading), or measures of uncertainty (implied volatility), or inquire about beliefs.

The only interview-based study in our review is by Fortin and Héroux (2023), who conduct seven interviews with investment managers, financial analysts, and chief information security officers at financial institutions. These users consider cyber risk disclosure as mostly useless for sophisticated capital market participants due to it being perceived as boilerplate, unreliable, and incomparable across firms. While getting first-hand users' impressions about the usefulness and informativeness of cyber risk disclosures is valuable, we caution that the sample of interviewees in this study is particularly small compared to best practices in interview-based research that call for at least 25–30 interviews to ensure a comprehensive range of views (Dai et al., 2019).

Cost of debt relates positively with the length, litigiousness, and lower readability of cyber risk disclosures (Harris et al., 2023), but negatively with the specificity of this disclosure about cyber investments (Havakhor et al., 2024). The latter also relates negatively to the cost of equity (Havakhor et al., 2024).

Long and even generic/boilerplate disclosures about information security seem to decrease information asymmetry in the stock market, at least in the early years of cyber risk disclosure (Gordon et al., 2010). However, in recent years, there is at best mixed evidence about the market outcomes from cyber disclosure sentiment and readability (Harris et al., 2023; Tsang et al., 2023). Further, cyber risk exposure, as reflected in conference call discussions, does increase uncertainty in the stock market (Jamilov et al., 2023).

Nonprofessional Investors

This section synthesizes eight studies that use experimental methods to examine nonprofessional investors' attitudes toward investing in a firm that discloses cyber risks or incidents, depending on the type and characteristics of disclosure (Panel D of Table 4).

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Four studies examine how nonprofessional investors' willingness to invest (routinely defined as likelihood to invest and/or amount invested) varies with different forms of assurance on cybersecurity (e.g., management assertion or self-assessment, assurance report by audit firm, or critical audit matter in audit report) (Frank et al., 2019; Gauch & Quick, 2025; Godewatta et al., 2024; J. A. Huang & Murthy, 2024). Results generally indicate that nonprofessional investors are more willing to invest when firms provide or purchase assurance related to cybersecurity; however, critical audit matters on the cyber incident included in the audit report are not related to the willingness to invest (Godewatta et al., 2024).[7] A fourth related study, Bansal and Axelton (2024), find that cyber disclosure verifiability, which is related to the concept of assurance, is positively related to investors' intentions to invest following a data breach, as it mitigates the drop in the trust that the firm can protect data.

One study (Cheng & Walton, 2019) examines whether the source of the data breach disclosure matters for nonprofessional investor willingness to invest. They find investors are more likely to want to invest when the firm discloses the cyber breach rather than an external party. How timely this disclosure is, however, does not seem to matter for nonprofessional investors. We highlight this study in light of our discussion in Section 4 about the source of disclosure being a unique feature of cyber disclosure.

Other studies in this area examine the outcome of the firm framing cyber risk management as a strategic initiative (negatively related to willingness to invest), the CEO issuing an apology after the breach (positively related to willingness to invest) (Demek & Kaplan, 2023), the interaction between cyber disclosures by non-breached firms before or after an industry peer is breached (the subsequent disclosure is negatively related to willingness to invest) (Kelton & Pennington, 2020), disclosure that the firm carries cyber insurance (positive relation to willingness to invest, and separate from the effect of cyber assurance) (Gauch & Quick, 2025; J. A. Huang & Murthy, 2024), and cyber disclosure specificity more broadly (positive relation to willingness to invest after a data breach) (Bansal & Axelton, 2024).

Importantly, we note that, over time, this literature has moved from recruiting experiment participants on Amazon Mechanical Turk (Amazon MTurk) to CloudResearch and Prolific; the latter platforms are considered better in terms of participant engagement, attention, and overall quality (e.g., Albert & Smilek, 2023; Douglas et al., 2023). The quality of the data platform used has implications for the weight one should assign to the studies in this area and for how careful we should be about interpreting the results of earlier studies that use Amazon MTurk.

Subsequent Cyberattacks

Panel E of Table 4 presents the eight studies that examine the relationship between cyber disclosure characteristics and the probability of a subsequent data breach. This stream of the cybersecurity literature is important as one of the unique features of cyber disclosure is that

7. A quick search in Audit Analytics United States and Canada for critical audit matters in audit reports reveals a very small number of cyber-related critical audit matters. Therefore, although theoretically interesting, the findings in Godewatta et al. (2024) have little practical implications for now.

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hackers may learn from the disclosures that firms provide, which will result in higher subsequent costs for the firm due to subsequent data breaches. Extant economics-based disclosure theory does not take into account the indirect cost of providing information that helps one's adversaries, as these costs are different from indirect proprietary, litigation, or political costs.

Three studies examine the relation between cyber risk disclosure length and subsequent cyberattacks, with mixed results: Li et al. (2018) and Jiang (2024) find that longer cyber disclosures are associated with a higher probability of future cyberattacks, while Berkman et al. (2021) find the opposite relation. More easy-to-read cyber disclosures, however, relate positively with future breaches (Jiang, 2024). Cyber exposure, as inferred from conference calls and annual reports, is predictive of subsequent breaches or increased future cyber risk (Du et al., 2024; Florackis et al., 2023; Jamilov et al., 2023), suggesting that managers correctly anticipate the potential of their firm suffering cyber breaches.

Specificity along the lines of disclosure of mitigation and defense mechanisms is consistently negatively related to the likelihood of future breach (Berkman et al., 2021; Ettredge et al., 2018; Wang et al., 2013). We believe this set of results is particularly important in this stream of research as it suggests that disclosing the specific protections the firm puts in place against cyberattacks does not translate into adversaries being able to break through these protections, which is one of the main arguments managers put forward against disclosing cyber-related details.

Other Outcomes

Finally, we review eight other studies that examine a range of other outcomes such that it is difficult to consider that there is a critical mass of studies on each of these outcomes (Panel F of Table 4). We quickly summarize these studies. Two studies examine the link between cyber risk and incident disclosure characteristics and audit fees, finding inconsistent results (Calderon & Gao, 2021; Li et al., 2020). Jamilov et al. (2023) and X. Huang et al. (2024) find a negative relation between cyber risk exposure disclosed in conference calls and disclosure delay, respectively, and firm performance. Gay (2017) and Vo and Pham (2024) examine the relation between cyber disclosure and other corporate disclosures such as news released in the media and characteristics of voluntary management guidance. Lattanzio and Ma (2023) is the only study to examine the relation between cyber risk disclosure and the firm's innovation activities as measured via R&D investments, patenting activity, and returns to innovation; they find no statistically significant relation with R&D investments, but increased patenting activity (more patents, more citations, shorter patenting time), however, with negative returns to innovation. More research is needed on these outcomes to draw conclusions that generalize across different countries, time periods, and so on.

Takeaways

The academic literature on the outcomes of cyber disclosure characteristics focuses largely on stock market−related measures. While there is some market reaction associated with certain disclosure channels and characteristics (8-K; more specific cyber disclosures), by and large there is little or no stock market reaction associated with most

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characteristics of cyber incident and risk disclosure. In experimental studies, however, results show that more transparent cyber disclosures (higher specificity, verifiability, assurance) have a favorable impact on retail investors' perceptions of the firm and willingness to invest. The findings on other stock and debt market outcomes are more mixed. The findings on the relation between cyber disclosure characteristics and subsequent cyberattacks indicate that cyber disclosure could be used, to some extent, by hackers to glean additional insights into companies, which translates in future cyberattacks, but we do not see evidence that when firms are specific about the types of protections they employ, future cyberattacks are more likely.

7. FUTURE RESEARCH

In this section, we summarize and elaborate on four directions that future research could take, based on our analysis and discussion of prior research. We organize potential future research questions into those related to developments around the theoretical framework used to conceptualize cybersecurity disclosure and those related to empirical issues that, in our perspective, remain thus far outstanding.

What Novel Insights Can Be Gleaned About Cybersecurity Disclosure by Recognizing the Unique Adversarial Costs That Firms Face When Making Such Disclosure?

Our review suggests that future research on cybersecurity disclosure could delve into the unique adversarial costs associated with cybersecurity disclosure. While there has been significant attention on the determinants of such disclosure, the specific adversarial costs incurred by firms remain underexplored. For instance, how do firms balance the need for transparency against the potential for increased vulnerability due to detailed disclosures? Investigating this tension can reveal insights into the strategic decisions firms make to protect themselves while complying with disclosure requirements. Understanding these costs could inform better regulatory policies that support transparency without unduly increasing firms' exposure to cyber threats.

Moreover, our review proposes that future research would gain from recognizing the unique nature of cybersecurity disclosure and thus going beyond the theories used in financial or other nonfinancial disclosure research. Specifically, the unique nature of this type of disclosure is awarded by new disclosure costs (beyond proprietary, litigation, and political costs) in the form of criminal adversarial costs since this type of disclosure could provide actionable insights for criminal hackers. As such, criminal hackers are a specific stakeholder group, or player, with an interest in cybersecurity disclosure that companies aim at investors. For example, examining cybersecurity disclosure using the lens of game theory offers several advantages. Game theory provides a framework for modeling strategic interactions between rational decision-makers, including adversaries, who anticipate each other's actions and respond strategically to achieve their objectives. By analyzing the incentives, strategies, and potential outcomes of different disclosure strategies, game theory can help understand how adversaries (hackers working for criminal organizations or for states seeking secret data) might react under various conditions. This approach allows

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organizations to anticipate and mitigate the risks posed by adversaries seeking to exploit disclosed information. Furthermore, the gains to be derived from hackers against specific firms could be integrated into the analysis to derive and observe potential disclosure strategies. Specifically, future research could examine the conditions under which criminal adversarial costs dominate the benefits from disclosing cybersecurity information.

To address the misalignment between theoretical frameworks and the underlying context, we argue that it is important to consider other theoretical frameworks and perspectives that offer more complete insights into the motivations, behaviors, and tactics of adversaries seeking to exploit cybersecurity disclosure. An adversarial risk analysis perspective focuses specifically on assessing and managing risks in environments where malicious actors actively seek to exploit vulnerabilities and weaknesses. By incorporating adversarial perspectives into risk assessments and decision-making processes, adversarial risk analysis helps organizations anticipate and defend against cyber threats related to cybersecurity disclosure.

Similarly, the cybersecurity economics perspective examines the economic incentives, trade-offs, and dynamics influencing the behavior of adversaries. By analyzing the economics of cybersecurity disclosure, cybersecurity economics sheds light on the incentives for adversaries to exploit disclosed information and informs strategies to enhance cybersecurity resilience. Further, organizational resilience theory emphasizes building capabilities such as flexibility, redundancy, agility, and learning to enable organizations to respond to and recover from cybersecurity incidents while minimizing the impact on stakeholders. By incorporating resilience principles into cybersecurity disclosure practices, organizations can enhance their capacity to manage the cyberspace's strategic landscape and mitigate the risks arising from disclosure.

The impact of adversarial costs on the quality and content of cybersecurity disclosures warrants examination. Firms may opt to underreport or use ambiguous language to mitigate risks, affecting the informativeness of the disclosures. Future research could explore how adversarial costs shape the nature of disclosures across different industries and regulatory environments, shedding light on the strategies employed by firms to navigate these challenges.

How Are Different Actors Within the Firm Working Together to Decide What Cybersecurity Disclosure Gets Communicated Outside of the Firm?

While many determinants of cybersecurity disclosure have been examined, the internal dynamics of firms regarding these decisions remain a "black box." Understanding how executives and managers—such as chief information officers (CIOs), CEOs, and board members—collaborate to determine what information to disclose is crucial. This encompasses exploring risk appetite, confidence levels, power structures, and the influence of institutional investors and regulatory compliance.

Empirically, it appears that most prior research relies on quantitative research methods, with a strong emphasis on event studies and textual analysis. Such data cannot shed light on the inner workings of cybersecurity disclosure, which is shaped by the interactions and

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tensions between the several corporate players (CIO, CEO, chief security officers [CSO], chief legal officer, audit committee, risk management committee, board of directors, internal auditors, etc.). Future research could broaden the methodological span to include qualitative research methods, including ethnography and participant observation, interviews, and analysis of internal documents. Specific questions could thus be addressed; for example, how are the legal/regulatory requirements traded off against investors' demands? What are the perceived benefits of further disclosure and how do they compare with potential negative spillover effects as hackers get access to such information? How are different actors within the firm working together to execute on and govern cybersecurity, as well as to decide what disclosure gets communicated outside the firm? Investigating how firms assess the materiality of cyber events to trigger disclosure can provide valuable understanding.

Furthermore, prospective studies may better address information needs of certain stakeholders and chart an evolutionary trajectory of cybersecurity risk disclosure over distinct temporal intervals, assessing the potential influence exerted by the occurrence or absence of cyberattacks on this evolutionary path. A longitudinal analysis (Madnick et al., 2024) could unveil the evolving trends and patterns in disclosure practices, shedding light on how these practices adapt and transform in response to the evolving cyber threat landscape. Such an in-depth exploration would offer a robust foundation for understanding the adaptive nature of cybersecurity disclosure and inform future strategies for bolstering organizational resilience against cyber risks.

The interplay of motivational factors, including regulatory requirements, reputation management, and legal liabilities, as well as personal and/or team dynamics, could significantly shape these internal discussions, offering insights into the determinants of cybersecurity disclosure practices.

How Do Stakeholders Perceive and Make Use of Cybersecurity Disclosure?

Examining stakeholder perceptions and utilization of cybersecurity disclosure is essential for understanding its broader impact. Different types of stakeholders, such as financial analysts, investors, regulators, customers, peer firms, and suppliers, may have varying interpretations and uses for disclosed information. For example, Fortin and Héroux (2023) highlight that financial analysts often dismiss cybersecurity disclosures as boilerplate, considering cybersecurity risk a systemic issue. However, different sectors, such as health care and finance, may find such disclosures more critical due to the severe consequences of cyberattacks, such as loss of life or financial savings.

Future research could focus on how stakeholders from different sectors and roles perceive and react to cybersecurity disclosures. This includes exploring whether these disclosures influence their decision-making, trust levels, and risk assessments. Understanding these dynamics can help tailor disclosure practices to meet stakeholder needs effectively, enhancing the relevance and effectiveness of cybersecurity communication.

Behavioral and psychological aspects surrounding cybersecurity disclosure (Maalem Lahcen et al., 2020) present a rich area for exploration. Future research could delve into understanding the behavioral implications of cybersecurity disclosure on various stakeholders.

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Investigating how different types of disclosures influence decision-making, trust levels, and behavioral intentions among stakeholders, while accounting for individual privacy concerns and psychological biases, is crucial. Research investigating ethical dimensions such as user privacy, data protection, and the balance between transparency and confidentiality could guide ethical and responsible cybersecurity disclosure practices (Fleischman et al., 2023).

Moreover, broadening the research scope beyond traditional equity markets to encompass diverse domains, including debt markets (Harris et al., 2023; Iyer et al., 2020), cybersecurity markets, cyber insurance markets (Franke, 2017), supplier relationships, and innovation ecosystems, offers an opportunity for a more comprehensive understanding of cybersecurity disclosure. Delving into the evolution of these varied markets in response to enhanced cybersecurity disclosure might reveal significant insights into market maturation, correlations with improved disclosure practices, and diverse market responses to cybersecurity disclosure.

What Roles Do Different Disclosure Channels Play for Cyber Disclosure and How Do Cybersecurity Disclosure Characteristics Differ Across Different Channels?

Future research could explore the varying roles and effectiveness of different disclosure channels, such as proxy statements, 8-K reports, social media, and corporate websites. Gordon et al. (2024) is the first study in this direction, comparing 8-Ks with other channels for disclosing cyber incidents. Each channel may have distinct characteristics and impacts on how information is conveyed and perceived. For example, social media may offer immediacy but lack depth, while formal reports provide detailed analysis but may not be as timely.

Future research can explore a spectrum of qualitative characteristics, extending beyond syntactic features to encompass semantic dimensions (Tsang et al., 2023). The investigation into how these dimensions manifest across diverse disclosure outlets is crucial for a comprehensive understanding. Moreover, given the considerable discretion firms hold in determining the materiality of cybersecurity information (Andrew, Baker, Sheehan, et al., 2023), researchers should focus on the nuanced decision-making processes. Understanding what, where, when, and how much to disclose requires an exploration of the factors influencing these choices (Gordon et al., 2010). The recent revelation of inconsistencies in cyber incident disclosure outlets, as outlined in a professional report (Audit Analytics, 2022), opens a pathway for future research. The examination of how content and quality vary across these outlets, coupled with an emphasis on the impact of public reaction through sentiment analysis on social media, represents an intriguing avenue.

Investigating the effectiveness of these channels in conveying cybersecurity risks and incidents can provide insights into how firms can optimize their communication strategies. This includes assessing the informativeness and impact of disclosures across different channels and understanding how public reactions, captured through sentiment analysis, vary based on the medium used. Additionally, exploring cross-jurisdictional differences and the influence of regulatory frameworks on the quality and content of disclosures across channels can offer a comprehensive understanding of global practices in cybersecurity communication.

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8. CONCLUSION

Our study offers a comprehensive overview of the prevailing landscape concerning the disclosure of cybersecurity risks and incidents, drawing upon insights from the existing body of literature. The synthesis of findings from prior studies reveals a myriad of factors motivating companies to disclose cybersecurity information. The synthesis underscores the pervasive influence of regulatory, public, and institutional pressures, alongside the intricate dynamics within the corporate realm, encapsulated by the interplay of board governance and firm characteristics. Additionally, the distinct attributes defining a cyber incident and the nature of cybersecurity threats emerge as critical determinants that significantly impact the disclosure practices. Recognizing the complexity of these determinants is crucial for stakeholders, providing valuable insights for organizations seeking to enhance their cybersecurity disclosure strategies. The synthesis also confirms the alignment between cybersecurity disclosure informativeness and regulatory expectations. However, our synthesis highlights a disagreement between cybersecurity disclosure and market reactions and the reasons behind the lack (or lack of statistical significance) of market response to a cybersecurity incident disclosure.

Moreover, the synthesis exposes significant gaps in the adherence to regulatory expectations. Companies, irrespective of their geographical origin, grapple with challenges such as boilerplate language, a lack of uniqueness, reduced readability, and an inclination toward more litigious language. Moreover, the analysis across diverse contexts reveals a shared limitation that firms struggle to provide the necessary depth and completeness in their disclosure related to cyber incidents. This synthesis underscores the pressing need for a more uniform and detailed approach to cybersecurity disclosure, aligning with regulatory expectations.

Our analysis highlights the complementary nature of theory-based and non-theory-based papers in providing insights into cybersecurity disclosure practices. Moreover, our review underscores the importance of utilizing other theoretical perspectives, particularly those tailored to address the adversarial nature of cybersecurity, such as game theory and adversarial risk analysis, to better capture the complexities inherent in cybersecurity disclosure environments. Methodologically, there is a prevalent reliance on quantitative textual analysis methods, particularly word frequency assessments and event studies. Notably, annual reports, the 10-K filings, stand out as primary channels for disclosure, yet venues such as conference calls and social media remain relatively underexplored.

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